Solid State

INTRODUCTION

A solid is defined as that form of matter which possesses rigidity and hence possesses a definite shape and a definite volume. Unlike gases and liquids in which the molecules are free to move about and hence constitute fluid state, in a solids the constituent particles are not free to move but oscillate about their fixed positions.

CLASSIFICATION OF SOLIDS

Solids are broadly classified into two types crystalline solids and amorphous solids.
A crystalline solid is a substance whose constituent particles possess regular orderly arrangement e.g. Sodium chloride, sucrose, diamond etc.

An amorphous solid is a substance whose constituent particles do not possess a regular orderly arrangement e.g. glass, plastics, rubber, starch, and proteins. Though amorphous solids do not possess long range regularity, in some cases they may possess small regions of orderly arrangement. These crystalline parts of an otherwise amorphous solid are known as ***crystallites.***

An amorphous solid does not posses a sharp melting point. It undergoes liquefication over a broad range of temperature. The amorphous solid do not posses any characteristic heat of fusion. When an amorphous solid is cut with the help of sharp edged knife it results in an irregular cut.

Amorphous substances are also, sometimes, referred to as super cooled liquids because they posses disorderly arrangement like liquids. In fact many amorphous solids such as glass are capable flowing. Careful examination of the window panes of very old houses reveals that the panes are thicker at the bottom than at the top because the glass has flown under constant influence of gravity.

DISTINCTION BETWEEN CRYSTALLINE AND AMORPHOUS SOLIDS

Crystalline solids	Amorphous solids
1. The internal arrangement of particles is regular so they possess definite and regular geometry	1. The internal arrangement of particles is irregular. Thus they do not have any definite geometry.
2. They have sharp melting points	2. They do not have sharp melting points
3. There is regularity in the external form when crystals are formed	3. There is no regularity in the external form when amorphous solids are formed
4. Crystalline solids give a regular cut when cut with a sharp – edged knife	4. Amorphous solids give irregular cut.
5. They have characteristic heat of fusion.	5. They do not have characteristic heat of fusion.
6. Crystalline solids are rigid and their shape is not distorted by mild distorting forces	6. Amorphous solid are not very rigid. These can be distorted by bending or compressing forces.
7. Crystalline solids are regarded as true solids	7. Amorphous solids are regarded as super cooled liquids or pseudo solids
8. Crystalline solids are anisotropic. This implies that physical properties such as refractive index, conductivity, thermal expansion etc are different in different directions. This is due to orderly	8. Amorphous solids are isotropic in nature. This implies that various physical properties are same in all the directions. This is because of random arrangement of particles.

arrangement of particles	

USES OF AMORPHOUS SOLIDS

Amorphous solids such as glass and plastics are very important materials and are widely used in construction, house ware, laboratory ware etc. Amorphous silica is likely to be the best material for converting sunlight into electricity (photovoltaic). Another well known amorphous solid is rubber which is used in making tyres shoes soles etc.

SPACE LATTICE OR CRYSTAL LATTICE

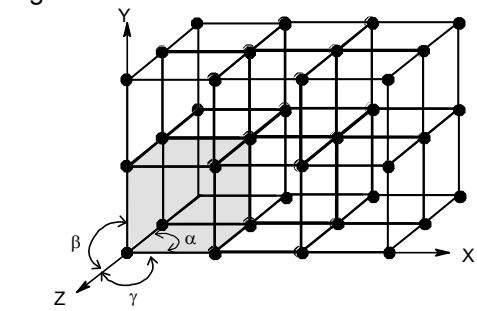
All crystals consists of regularly repeating array of atoms, molecules or ions which are the structural units (or basic units). It is much more convenient to represent each unit of pattern by a point, called lattice point, rather than drawing the entire unit of pattern. This results in a three dimensional orderly arrangement of points called a space lattice or a crystal lattice

Thus, a space lattice may be defined as a regular three dimensional arrangement of identical points in space or it can be defined as an array of points showing how molecules, atoms or ions are arranged at different sites in three dimensional space.

It must be noted that

(a) Each lattice point has the same environment as that of any other point in the lattice

(b) The constituent particles have always to be represented by a lattice point, irrespective of whether it contains a single atom or more than one atoms



A crystal or space lattice - The shaded portion represents the unit cell

UNIT CELL

A unit cell is the smallest repeating unit in space lattice which when repeated over and over again results in a crystal of the given substance. Unit cell may be also defined as a three dimensional group of lattice points that generates the whole lattice on repetition.

BRAVAIS LATTICES

The French crystallographer August Bravais in 1848 showed from geometrical consideration that there can be only 14 different ways in which similar points can be arranged in a three dimensional space. Thus the total no. of space lattices belonging to all the seven basic crystal system but together is only 14.

Types of unit cells

1. **Simple unit cell** having lattice points only at the corners is called simple, primitive or basic unit cell. A crystal lattice having primitive unit cell is called simple crystal lattice
2. **Face centred cubic lattice** (fcc) – A unit cell in which the lattice point is at the centre of each face as well as at the corner.
3. **Body centred cubic lattice** (bcc) A unit cell having a lattice point at the centre of the body as well as at the corners.

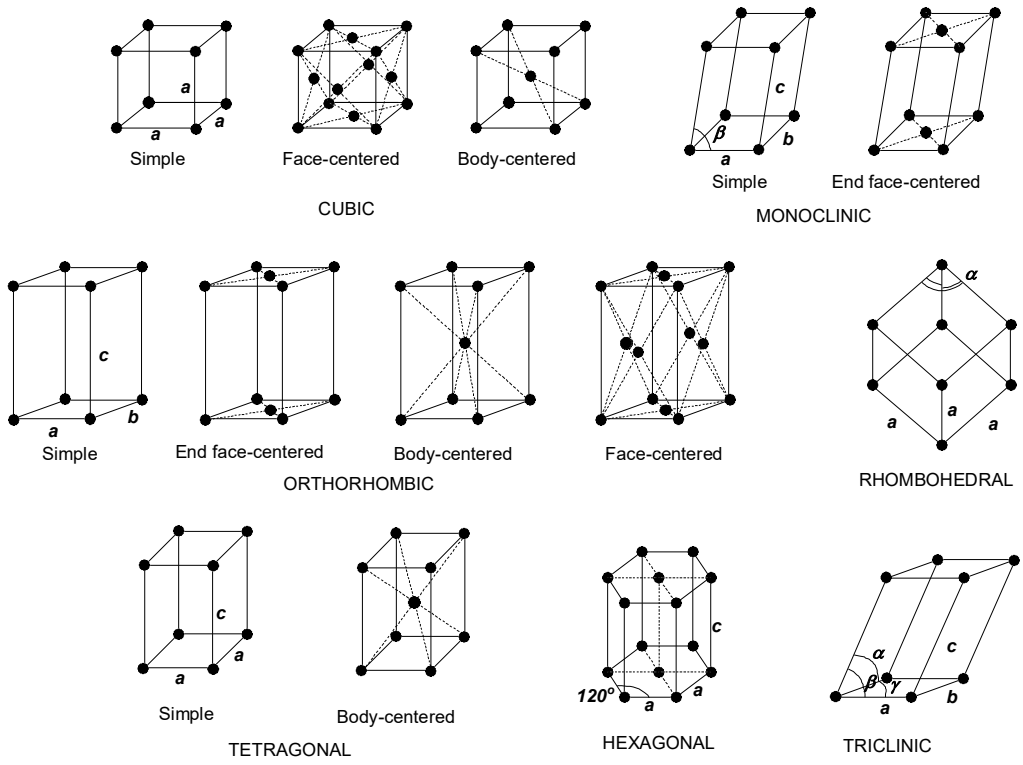
Another type of unit cell, called end – centred unit cell is possible for orthorhombic and monoclinic crystal types.

In an end centred there are lattice points in the face centres of only one set of faces in addition to the lattice pints at the corners of the unit cell

The various types of unit cells possible for different crystal classes (in all seven) are given below in tabular form

Crystals class	Axial distances	Angles	Possible types of unit cells	Examples
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$	Primitive, Body centred face centred	Copper , KCl, NaCl zinc blende, diamond
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	Primitive, body centred	SnO_2, White tin, TiO_2
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	Primitive body centred, face centred end centred	Rhombic sulphur, KNO_3, $CaCO_3$
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°$ $\gamma = 120°$	Primitive	Graphite, Mg, ZnO
Trigonal or Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	Primitive	($CaCO_3$) Calcite, HgS(Cinnabar)
Monoclinic	$a \neq b \neq c$	$\alpha = \beta = 90°$ $\gamma \neq 90°$	Primitive and end centred	Monoclinic sulphur, $Na_2SO_4.10H_2O$
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	Primitive	$K_2Cr_2O_7$, $CuSO_4.5H_2O$

The Bravais space lattices associated with various crystal system are show in fig below



CALCULATION OF NUMBER OF PARTICLES PER UNIT CELL

The no of atom in a unit cell can be calculated by keeping in view following points

1. An atom at the corners is shared by eight unit cells. Hence the contribution of an atom at the corner to a particular cell $= \frac{1}{8}$
2. An atom at the face is shared by two unit cells. Hence the contribution of an atom at the face to a particular cell = $\frac{1}{2}$
3. An atom at the edge centre is shared by four unit cells in the lattice and hence contributes only $\frac{1}{4}$ to a particular unit cell.
4. An atom at the body centre of a unit cell belongs entirely to it, so its contribution = 1

The number of atoms per unit cell is in the same ratio as the stoichiometry of the compound. Hence it helps to predict the formula of the compound

SIMPLE CUBIC LATTICE

There are eight atoms at the corners. Each corner atom makes 1/8 contribution to the unit cell.

$\therefore$ No. of atoms present in the unit cell = $\frac{1}{8} \times 8 = 1$

Exercise 1.

How many atoms are present in a primitive cell?

BODY CENTRED CUBIC (BCC)

BCC has 8 atoms at the corners and one atom, within the body. Each corner atom makes 1/8 contribution and the contribution of atom within the body = 1

$\therefore$ No of atoms present in bcc = $\frac{1}{8} \times 8$ (at corner) + 1(at the body centre)

= 1+1 =2

FACE CENTRED CUBIC (FCC)

Fcc has 8 atoms at the corners and 6 atoms on the faces (one on each face)

Contribution by atoms at the corners = $\frac{1}{8} \times 8 = 1$

Contribution by atom on the face = $\frac{1}{2} \times 6 = 3$

$\therefore$ Number of atoms present in fcc unit cell = 1+3 = 4

Illustration 1. *A compound formed by elements A and B has a cubic structure in which A atoms are at the corners of the cube and B atoms are at face centres. Derive the formula of the compound.*

Solution: As 'A' atom are present at the 8 corners of the cube therefore no of atoms of A in the unit cell = $\frac{1}{8} \times 8 = 1$

As B atoms present at the face centres of the cube, therefore no of atoms of B in the unit cell = $\frac{1}{2} \times 6 = 3$

Hence the formula of compound is AB_3.

Illustration 2. *Potassium crystallizes in a body centred cubic lattice. What is the approximate number of until cells in 4.0 g of potassium? Atomic mass of potassium = 39.*

***Solution*:** In bcc unit cell there are 8 atoms at the corners of the cube and one atom at the body centre

$\therefore$ No of atoms per unit = $8 \times \frac{1}{8} + 1 = 2$

No of atoms is 4.0 g of potassium = $\frac{4}{39} \times 6.023 \times 10^{23}$

$\therefore$ No of unit cells in 4.0 g potassium = $\frac{4}{39} \times \frac{6.023 \times 10^{23}}{2} = 3.09 \times 10^{22}$

Illustration 3. *A compound made up of A and B atoms have a crystalline structure, in which A forms Hexagonal close packed structure and B occupies 2/3 of octahedral holes. What will be the simplest molecular formula?*

Solution: Effective number of A atoms forming HCP = 6
Effective number of octahedral holes in HCP = 6
So molecular formula will
$A_6B_{6\times 2/3} = A_6B_4 = A_3B_2$

Illustration 4. *An ionic compound made up of atoms A and B has a face-centred cubic arrangement in which atoms A are at the corners and atoms B are at the face-centres. If one of the atoms is missing from the corner, what is the simplest formula of the compound?*

Solution: Number of atoms of A at the corners = 7 (because one A is missing)

$\therefore$ Contribution atoms of A towards unit cell $= 7 \times \frac{1}{8} = \frac{7}{8}$

Number of atoms B at face-centres = 6

$\therefore$ Contribution of atom B towards unit cell $= 6 \times \frac{1}{2} = 3$

Ratio of A : B $= \frac{7}{8} : 3 = 7 : 24$

$\therefore$ Formula is A_7B_{24}

Illustration 5. *In a face centered cubic arrangement of X and Y atoms, whose Y atoms are at the corner of the unit cell and X-atoms at the face centers. One of the X-atoms is missing from one of the faces in the unit cell. The simplest formula of the compound is*

(A) X_5Y_2 *(B)* X_2Y_5

(C) XY_3 *(D)* X_3Y

Solution: **(A)**

Illustration 6. *A solid crystal is composed of X, Y and Z atoms. Y atoms are occupying 50% of octahedral voids, where as X atoms are occupying the 100% tetrahedral void with Z atoms in ccp array arrangement, then the rational formula of the compound in the given crystal is*

(A) $X_8Y_2Z_4$ *(B)* $X_5Y_{10}Z_8$

(C) X_4YZ_2 *(D)* $X_{16}Y_4Z_8$

Solution: **(A)**

Illustration 7. *In FCC lattice of NaCl structure, if the diameter of* Na^+ *is x, and the radius of* Cl^- *is y, then the bond length of NaCl in the crystal is*

(A) 2x + 2y *(B) x + y*

(C) $\frac{x}{2} + \frac{y}{2}$ *(D) None*

Solution: **(C)**

Illustration 8. *In a cubic packed structure of mixed oxides, the lattice is made up of oxide ions, one fifth of tetrahedral voids are occupied by divalent (*X^{++}*) ions, while one-half of the octahedral voids are occupied by trivalent ions (*Y^{+3}*), then the formula of the oxide is.*

(A) XY_2O_4 *(B)* X_2YO_4

(C) $X_4Y_5O_{10}$ *(D)* $X_5Y_4O_{10}$

Solution: **(C)** In ccp anions occupy primitives of the cube while cations occupied voids. In ccp there are two tetrahedral voids and one octahedral holes per anion. For one oxygen atom there are two tetrahedral holes and one octahedral hole.

Since one fifth of the tetrahedral voids are occupied by divalent cations (X^{2+})

$\therefore$ number of divalent cations in tetrahedral voids $= 2 \times \frac{1}{5}$.

Since half of the octahedral voids are occupied by trivalent cations (Y^{3+})

$\therefore$ number of trivalent cations $= 1 \times \frac{1}{2}$.

So the formula is the compounds is $(X)_{2\times\frac{1}{5}}\ (Y)_{\frac{1}{2}}\ (O)_1$

or $X_{\frac{2}{5}}Y_{\frac{1}{2}}O_1$,

or $\mathbf{X_4Y_5O_{10}}$

Illustration 9. *If three elements X, Y & Z crystallized in cubic solid lattice with X atoms at corners, Y atoms at cube centre & Z-atoms at the edges, then the formula of the compound is:*

(A) XYZ *(B) XY_3Z*

(C) XYZ_3 *(D) X_3YZ*

Solution: **(C)** Atom X is shared by 8 corners

1 atoms of Z is shared by 4 unit cells

Atom Y is present at centre of the unit cell

Hence, effective number of atoms of X per unit cell = $8 \times \frac{1}{8} = 1$

Effective number of atoms of Z per unit cell = $\frac{12}{4} = 3$

Effective number of atoms of Y per unit cell = 1

Hence, the formula of the compound is $\mathbf{XYZ_3}$

Exercise 2.

(i) A compound formed by elements X and Y crystallizes in the cubic structure where X atoms are at the corners of the cube and Y atoms at the alternate faces. What is the formula of the compound?

(ii) Metallic gold crystallizes in the face centred cubic lattice. What is the approximate number of unit cells in 2.0 g of gold? Atomic mass of gold is 197.

CLOSE PACKING IN CRYSTALS

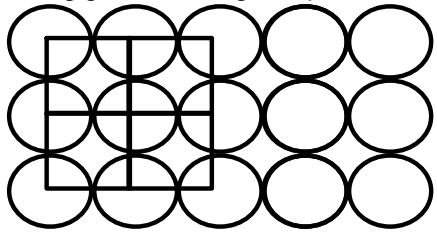
In order to understand the packing of the constituent particles in a crystal, it is assumed that these particles are hard spheres of identical size (eg those of metal). The packing of these spheres takes place in such a way that they occupy the maximum available space and hence the crystal has maximum density. This type of packing is called close packing.

TWO DIMENSIONAL PACKING

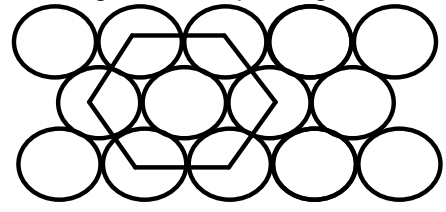
When the rows are combined touching each other, the crystal plane is obtained. The rows can be combined in two different ways

(i) The particles when placed in the adjacent rows, show a horizontal as well as vertical alignment and form squares. This type of packing is called square close packing

(ii) The particles in every next row are placed in the depressions between the particles of the first row. The particles in the third row will be vertically aligned with those in the first row. This type of packing gives a hexagonal pattern and is called hexagonal close packing



Square close packing (coordination no 4)

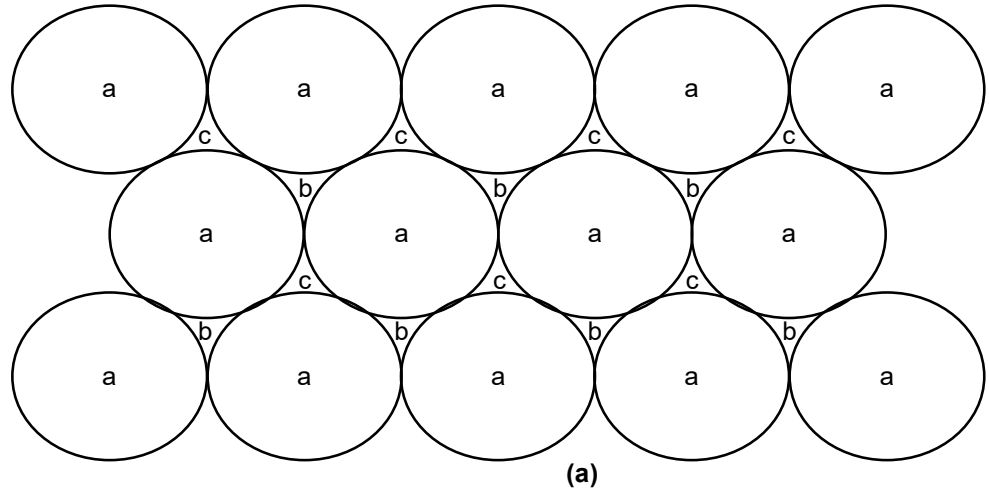


Hexagonal close packing(coordination no 6)

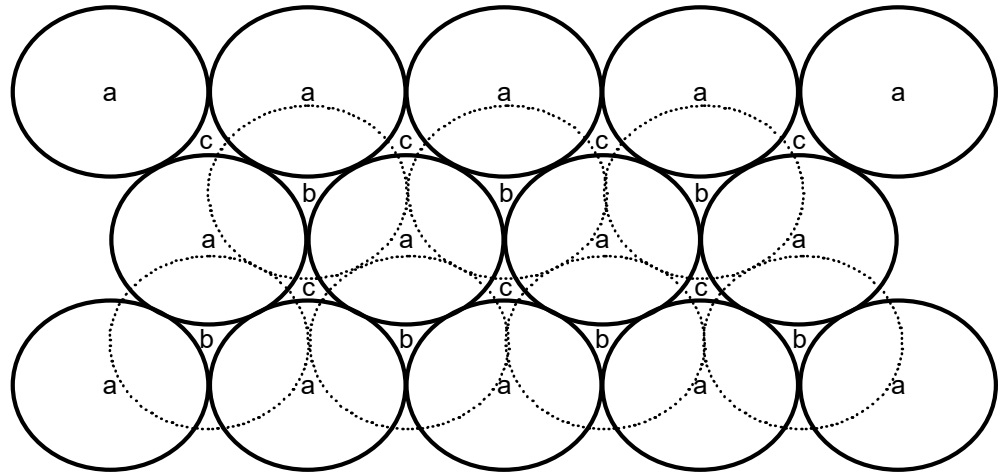
THREE DIMENSIONAL PACKING

In two dimensional packing, a more efficient packing is given by hexagonal close packing. In order to develop three dimensional close packing let us retain the hexagonal close packing in the first layer. If the spheres in the second layer are just placed over the spheres in the first layer so that the spheres in the two layers are vertically aligned, its voids will come above the voids in the first layer. This is an inefficient way of filling the space.

When the second layer is placed in such a way that its spheres find place in the 'b' voids of the first layer, the 'c' voids will be left unoccupied. Since under this arrangement no sphere can be placed in them, (c voids), i.e. only half the triangular voids in the first layer are occupied by spheres in the second layer (i.e. either b or c)



(a)



(b)

There are two alternative ways in which spheres in the third layer can be arranged over the second layer

(1) When a third layer is placed over the second layer in such a way that the spheres cover the tetrahedral or 'a' voids; a three dimensional closest packing is obtained where the spheres in every third or alternate layers are vertically aligned (i.e. the third layer is directly above the first, the fourth above the second layer and so on) calling the first layer A and second layer as layer B, the arrangement is called ABAB ……. pattern or hexagonal close packing (hcp) as it has hexagonal symmetry.

(2) When a third layer is placed over the second layer in such a way that spheres cover the octahedral or 'c' voids, a layer different from layers A and B is produced. Let it be layer 'C'. Continuing further a packing is obtained where the spheres in every fourth layer will be vertically aligned to the spheres present in the first layer. This pattern of stacking spheres is called ABCABC …….. pattern or cubic close packing (ccp). It is similar to face centred cubic (fcc) packing as it has cubic symmetry

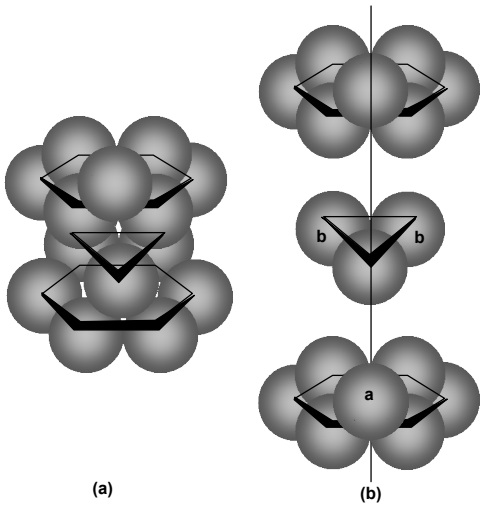


Fig. (X) Hexagonal closest packing of spheres: (a) normal and (b) exploded view

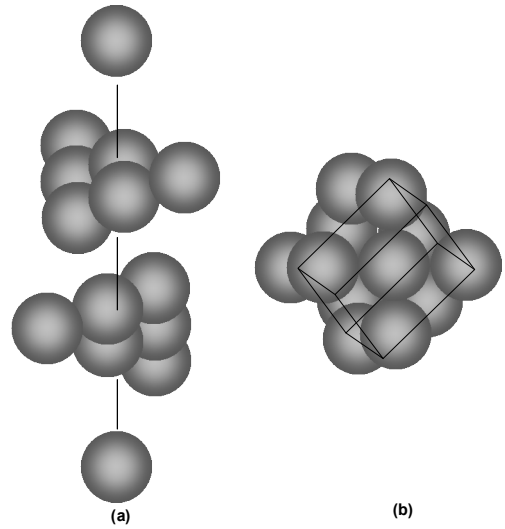


Fig. (X): Cubic closest packing of spheres: (a) generation of unit from closest–packed layers, and (b) rotation to show cubic symmetry

In both hcp and ccp methods of stacking, a sphere is in contact with 6 other spheres in its own layer. It directly touches 3 spheres in the layer above and three spheres in the layer below. Thus sphere has 12 close neighbours. The number of nearest neighbours in a packing is called coordination number. In close packing arrangement (hcp & ccp) each sphere has a coordination number of 12.

Exercise 3.

What is the coordination number of the atoms arranged in (ABC ABC ABC…) type structure?

PACKING FRACTIONS

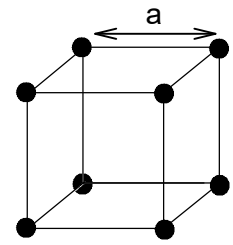
Both of the above patterns of packing (i.e. hcp & ccp) though different in form are equally efficient. They occupy the maximum possible space which is about 74% of the available volume. Hence they are called closest packing.

In addition to the above two types of arrangements a third type of arrangement found in metals is body centred cubic (bcc) in which space occupied is about 68%.

CALCULATION OF THE SPACE OCCUPIED

In simple cubic unit cell

Let ‘a’ be the edge length of the unit cell and r be the radius of sphere.



As sphere are touching each other

Therefore a = 2r

No. of spheres per unit cell = $\frac{1}{8}\times 8 = 1$

Volume of the sphere = $\frac{4}{3}\pi r^3$

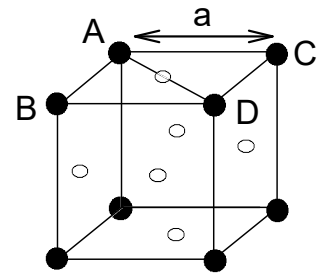
Volume of the cube = $a^3 = (2r)^3 = 8r^3$

$\therefore$ Fraction of the space occupied = $\frac{4/3\pi r^3}{8r^3} = 0.524$

$\therefore$ % occupied = 52.4 %

In face centred unit cell

Let ‘r’ be the radius of sphere and ‘a’ be the edge length of the cube



As there are 4 sphere in fcc unit cell

$\therefore$ Volume of four spheres $= 4\left(\frac{4}{3}\pi r^3\right)$

In fcc, the corner spheres are in touch with the face centred sphere. Therefore, face diagonal AD is equal to four times the radius of sphere

AD = 4r

But from the right angled triangle ACD

AD= $\sqrt{AC^2 + DC^2} = \sqrt{a^2 + a^2} = \sqrt{2}\,a$

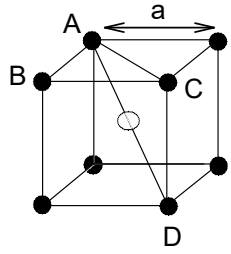
$\sqrt{2}\,a = 4r$ or $a = \frac{4}{\sqrt{2}}r$

$\therefore$ volume of cube = $\left(\frac{4}{\sqrt{2}}r\right)^3 = \frac{64}{2\sqrt{2}}r^3$

percentage of space occupied by sphere

$= \frac{\text{volume of sphere}}{\text{volume of cube}}\times 100 = \frac{\frac{16}{3}\pi r^3}{\frac{64}{2\sqrt{2}}r^3}\times 100 = 74\%$

In body centred cubic unit cell



Let ‘r’ be the radius of sphere and ‘a’ be the edge length of the cube

As the sphere at the centre touches the sphere at the corner. Therefore body diagonal AD = 4r

Face diagonal AC = $\sqrt{AB^2 + BC^2} = \sqrt{a^2 + a^2} = \sqrt{2}\,a$

In right angled triangle ACD =

$AD = \sqrt{AC^2 + CD^2} = \sqrt{2a^2 + a^2} = \sqrt{3}\,a$

$\sqrt{3}\,a = 4r$

$a = \dfrac{4r}{\sqrt{3}}$

$\therefore$ Volume of the unit cell = $a^3 = \left(\dfrac{4r}{\sqrt{3}}\right)^3 = \dfrac{64r^3}{3\sqrt{3}}$

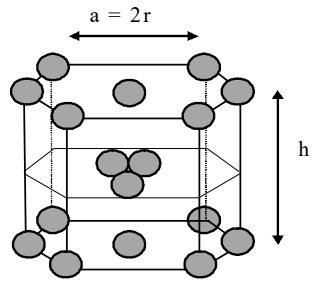
No. of spheres in bcc = 2

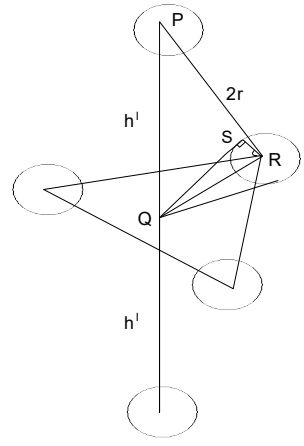
$\therefore$ volume of 2 spheres = $2 \times 4/3\pi r^3$

$\therefore$ percentage of space occupied by spheres

$$= \frac{\text{volume of sphere}}{\text{volume of cube}} \times 100 = \frac{\frac{8}{3}\pi r^3 \times 100}{\frac{64\ r^3}{3\sqrt{3}}} = \frac{8}{3} \times \frac{22}{7} \times \frac{3\sqrt{3}}{64} = 68\%$$

Illustration 10. *Show by simple calculation that the percentage of space occupied by spheres in hexagonal cubic packing (hcp) is 74%*





Solution: Let the edge of hexagonal base =a

And the height of hexagon = h

And radius of sphere = r

The centre sphere of the first layer lies exactly over the void of 2nd layer B.

The centre sphere and the spheres of 2nd layer B are in touch

So, In Δ PQR (an equilateral triangle)

PR = 2r, Draw QS tangent at points

$\therefore$ In Δ QRS $\quad \angle QRS = 30°, \quad SR = r$

$\text{Cos}30° = \dfrac{SR}{QR}$

$QR = \dfrac{r}{\sqrt{3}/2} = \dfrac{2r}{\sqrt{3}}$

$\therefore\ PQ = \sqrt{PR^2 - QR^2} = \sqrt{4r^2 - \dfrac{4r^2}{3}}$

$h^1 = \sqrt{\dfrac{8r^2}{3}} = 2\sqrt{\dfrac{2}{3}}r$

$\therefore$ h = $2h^1 = 4\sqrt{\frac{2}{3}}r$

Now, volume of hexagon = area of base x height

= $\frac{6\times\sqrt{3}}{4}a^2\times h \Rightarrow 6\times\frac{\sqrt{3}}{4}(2r)^2\times 4\sqrt{\frac{2}{3}}r$

[$\therefore$Area of hexagonal can be divided into six equilateral triangle with side 2r)

No. of sphere in hcp = $12\times\frac{1}{6}+\frac{1}{2}\times 2+3$

= 2+1+3 = 6

$\therefore$ Volume of spheres = $6\times 4/3\pi r^3$

$\therefore$ Percentage of space occupied by sphere =

$$\frac{6\times\frac{4}{3}\pi r^3}{6\times\frac{\sqrt{3}}{4}\times 4r^2\times 4\sqrt{\frac{2}{3}}r}\times 100 = 74\%$$

Illustration 11. *An element crystallizes into a structure which may be described by a cubic type of unit cell having one atom in each corner of the cube and two atoms on one of its face diagonals. If the volume of this unit cell is 24 x $10^{-24}cm^3$ and density of the element is 7.20gm/cm^3, calculate no. of atoms present in 200gm of the element.*

Solution: Number of atoms contributed in one unit cell

= one atom from the eight corners + one atom from the two face diagonals

= 1+1 = 2 atoms

Mass of one unit vell = volume $\times$ its density

= 24×10^{-24} $cm^3\times 7.2$ gm cm^3

= 172.8×10^{-24} gm

$\therefore$ 172.8 10^{-24} gm is the mass of one – unit cell i.e., 2 atoms

$\therefore$ 200 gm is the mass = $\frac{2\times 200}{172.8\times 10^{-24}}$ atoms **= 2.3148 $\times$ 10^{24} atoms**

Illustration 12. *Calculate the void fraction for the structure formed by A and B atoms such that A form hexagonal closed packed structure and B occupies 2/3 of octahedral voids. Assuming that B atoms exactly fitting into octahedral voids in the HCP formed by A*

Solution: Total volume of A atom = $\frac{6\times 4}{3}\pi r_A^{\ 3}$

Total volume of B atoms = $4\times\frac{4}{3}\pi r_B^3 = 4\times\frac{4}{3}\pi(0.414r_A)^3$

Since $\frac{r_B}{r_A} = 0.414$ as B is in octahedral void of A

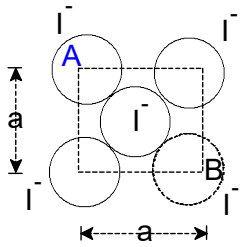
Volume of HCP = $24\sqrt{2}r_A^3$

Packing fraction = $\frac{6\times\frac{4}{3}\pi r_A^3+4\times\frac{4}{3}\pi(0.414r_A)^3}{24\sqrt{2}r_A^3}$ = 0.7756

Void fraction = 1-0.7756 = 0.2244

Illustration 13. *Lithium iodide crystal has a face-centred cubic unit cell. If the edge length of the unit cell is 620 pm, determine the ionic radius of I^- ion.*

Solution: As LiI has fcc arrangement, I^- ions will occupy the corners and face-centres. These ions will touch each other along the face diagonal.



$\therefore$ Face diagonal AB $= 4r_1^- = \sqrt{a^2 + a^2} = \sqrt{2}a$

$\therefore \quad r_1 = \dfrac{\sqrt{2}a}{4} = \dfrac{1.414 \times 620 \text{ pm}}{4} = 219.17 \text{ pm}$

Illustration 14. *When heated above 916°C, iron changes its crystal structure from bcc to ccp structure without any change in the radius of atom. The ratio of density of the crystal before heating and after heating is:*

(A) 1.089 *(B) 0.918*

(C) 0.725 *(D) 1.231*

Solution: **(B)** $\dfrac{d_{bcc}}{d_{ccp}} = \dfrac{\text{packing efficiency of bcc}}{\text{packing efficiency of ccp}} = \dfrac{67.92}{74.02} = 0.918$

Illustration 15. *Select the correct statements:-*

(A) For CsCl unit cell (edge-length = a), $r_c + r_a = \dfrac{\sqrt{3}}{2}a$

(B) For NaCl unit cell (edge-length = ℓ), $r_c + r_a = \dfrac{\ell}{2}$

(C) The void space in a b.c.c. unit cell is 0.68

(D) The void space % in a face-centered unit cell is 26%

Solution: **(A), (B), (C), (D).** In bcc structure are $r_+ + r_- = \dfrac{\sqrt{3}}{2}a$

Hence, for CsCl, $r_C + r_a = \dfrac{\sqrt{3}}{2}\mathbf{a}$

$\therefore$ (A) is correct

Since, NaCl crystalise in fcc structure

$\therefore$ $2r_C + 2r_a$ = edge length of the unit cell

Hence, $r_C + r_a = \dfrac{\boldsymbol{l}}{\mathbf{2}}$

$\therefore$ (B) is correct

Since packing fraction of a bcc unit cell is 0.68

$\therefore$ void space = 1–0.68 = **0.32**

$\therefore$(C) is incorrect

In fcc unit cell PF = 74%

$\therefore$ VF = 100–74 = **26%**

$\therefore$ (D) is correct

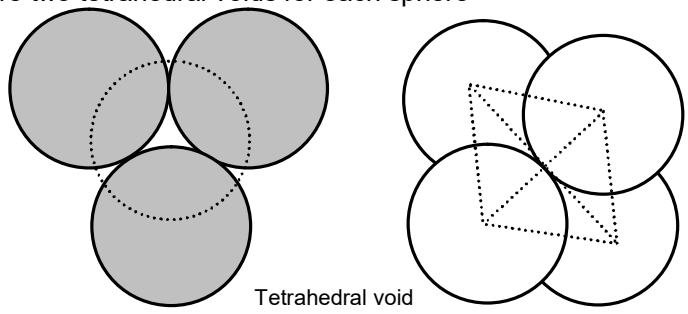
Exercise 4.
(a) What is the empty fraction in case of S.C., B.C.C and F.C.C?
(b) Which is most packed structure S.C., B.C.C or F.C.C?

INTERSTITIAL VOIDS

In hcp as well as ccp only 74% of the available space is occupied by spheres. The remaining space is vacant and constitutes interstitial voids or interstices or holes. These are of two types
(a) Tetrahedral voids (b) Octahedral voids

Tetrahedral voids

In close packing arrangement, each sphere in the second layer rests on the hollow (triangular void) in three touching spheres in the first layer. The centres of theses four spheres are at the corners of a regular tetrahedral. The vacant space between these four touching spheres is called tetrahedral void. In a close packing, the number of tetrahedral void is double the number of spheres, so there are two tetrahedral voids for each sphere



Tetrahedral void

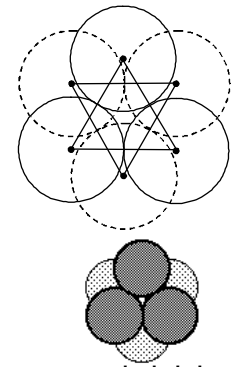
Radius of the tetrahedral void relative to the radius of the sphere is 0.225

i.e. $\frac{r_{void}}{r_{sphere}} = 0.225$

In a multi layered close packed structure , there is a tetrahedral hole above and below each atom hence there is twice as many tetrahedral holes as there are in close packed atoms

Octahedral voids

As already discussed the spheres in the second layer rest on the triangular voids in the first layer. However, one half of the triangular voids in the first layer are occupied by spheres in the second layer while the other half remains unoccupied. The triangular voids 'b' in the first layer are overlapped by the triangular voids in the second layer. The interstitial void, formed by combination of two triangular voids of the first and second layer is called octahedral void because this is enclosed between six spheres centres of which occupy corners of a regular octahedron



Octahedral void

In close packing, the number of octahedral voids is equal to the number of spheres. Thus, there is only one octahedral void associated with each sphere. Radius of the octahedral void in relation to the radius of the sphere is 0.414 i.e. $\frac{r_{Void}}{r_{sphere}} = 0.414$

LOCATING TETRAHEDRAL AND OCTAHEDRAL VOIDS

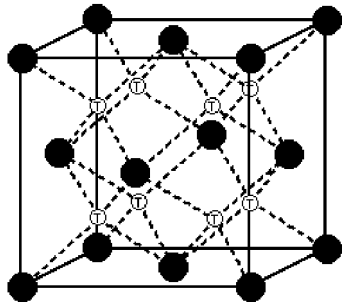
The close packed structures have both octahedral and tetrahedral voids. In a ccp structure, there is 1 octahedral void in the centre of the body and 12 octahedral void on the edges. Each one of which is common to four other unit cells. Thus, in cubic close packed structure.

Octahedral voids in the centre of the cube =1

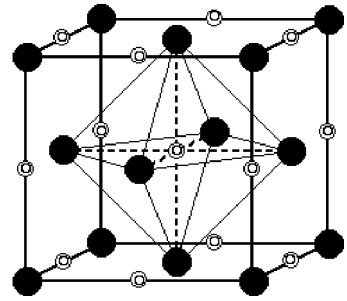
Effective number of octahedral voids located at the 12 edge of = $12 \times \frac{1}{4} = 3$

$\therefore$ Total number of octahedral voids = 4

In ccp structure, there are 8 tetrahedral voids. In close packed structure, there are eight spheres in the corners of the unit cell and each sphere is in contact with three groups giving rise to eight tetrahedral voids



Circles labelled T represent the centers of the tetrahedral interstices in the ccp arrangement of anions. The unit cell "owns" 8 tetrahedral sites.



Circles labelled O represent centers of the octahedral interstices in the ccp arrangement of anions (fcc unit cell). The cell "owns" 4 octahedral sites.

Illustration 16. *In a solid, oxide ions are arranged in ccp. Cations A occupy one – sixth of the tetrahedral voids and cations B occupy one third of the octahedral voids. What is the formula of the compound?*

Solution: In ccp with each oxide there would be 2 tetrahedral voids and one octahedral voids 1/3rd octahedral voids is occupied by B and 1/6th tetrahedral void by A. Therefore the compound can be

$A_{2/6}B_{1/3}O$ or ABO_3

Illustration 17. *In a crystalline solid, having formula AB_2O_4, oxide ions are arranged in cubic close packed lattice while cations A are present in tetrahedral voids and cations B are present in octahedral voids*

(i) What percentage of the tetrahedral voids is occupied by A?

(ii) What percentage of the octahedral voids is occupied by B?

Solution: In a cubic close packed lattice of oxide ions there would be two tetrahedral voids and one octahedral void for each oxide ion.

$\therefore$ For four oxide ions there would be 8 tetrahedral and four octahedral voids two are occupied by B.

Percentage of tetrahedral voids occupied by A = $\frac{1}{8} \times 100 = 12.5\%$

Percentage of tetrahedral voids occupied by B = $\frac{2}{4} \times 100 = 50\%$

Illustration 18. *A binary solid has zinc blend structure with* B^- *ions constituting the lattice and* A^+ *ions occupying 25% tetrahedral voids. The formula of solid is*

(A) AB *(B)* A_2B

(C) AB_2 *(D)* AB_4

Solution: **(C)** No. of B^- ions in unit cell = $8\times\frac{1}{8}+\frac{1}{2}\times 6=4$

Now A^+ ion occupies 25% of tetrahedral voids

$\therefore$ No. of $A^+=\frac{8\times 25}{100}=2$

Thus ratio of A^+ to B^- is 1:2

Hence formula is AB_2

Illustration 19. *A compound crystallises as follows:*

Ions "A" are at corners of a cubic unit cell and "B" ions at face centres of a cubic unit cell and "C" ions in 1/4th of the total tetrahedral void. Assuming if this is dissolved, only the ions in the tetrahedral voids are dissociated completely in water, which one of the following statements is true. (Assuming all are univalent ions) and also A and C are cations and B is an anions.

(A) Boiling point of same concentration of $AlCl_3$ *solution (100% dissociation) will be greater than that of this solution.*

(B) Boiling point of same concentration of $Ca_3(PO_4)_2$ *(100% dissociation) will be greater than that of this solution.*

(C) Boiling point of same concentration of sucrose will be greater than this solution

(D) Data insufficient to predict.

Solution: **(B)** Compound is AB_3C_2

So if C alone dissociates

$AB_3C_2 \longrightarrow [AB_3]^{2-}+2[C]^+$

Total 3 ions

Exercise 5.

(a) In a crystalline solid having molecular formula A_2B anion B are arranged in cubic close packed lattice and cations A are equally distributed between octahedral and tetrahedral voids

(i) What percentage of octahedral voids is occupied?

(ii) What percentage of tetrahedral voids is occupied?

(b) In a compound, oxide ions have ccp arrangement cations A are present in 1/8th of the tetrahedral voids and cations B occupy half the octahedral voids. What is the simplest formula of the compound?

Exercise 6.

(i) What is the number of octahedral voids in case of H.C.P and F.C.C?

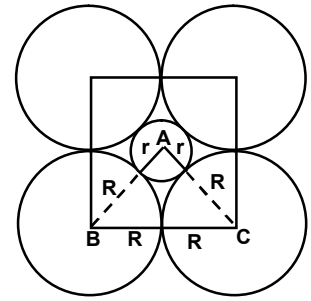
(ii) True or False

Number of tetrahedral voids is a whole number multiple of the number of octahedral void where n is any integer.

SIZES OF TETRAHEDRAL AND OCTAHEDRAL VOIDS

(i) Derivation of the relationship between the radius (r) of the octahedral void and the radius (R) of the atoms in close packing.

A sphere into the octahedral void is shown in the diagram. A sphere above and a sphere below this small sphere have not been shown in the figure. ABC is a right angled triangle. The centre of void is A.



Applying Pythagoras theorem.

$BC^2 = AB^2 + AC^2$

$$(2R)^2 = (R+r)^2 + (R+r)^2 = 2(R+r)^2 \Rightarrow \frac{4R^2}{2} = (R+r)^2$$

$$\Rightarrow \left(\sqrt{2}\right) = (R+r)^2$$

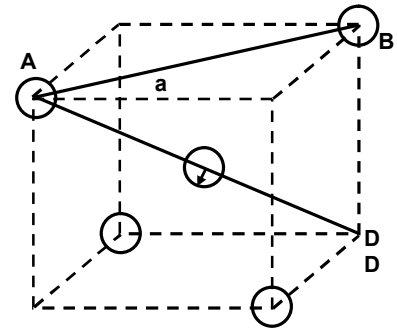
$$\Rightarrow \sqrt{2}R = R + r$$

$$\Rightarrow r = \sqrt{2}R - R = (1.414 - 1)R$$

r = 0.414 R

(ii) Derivation of the relationship between radius (r) of the tetrahedral void and the radius (R) of the atoms in close packing: To simplify calculations, a tetrahedral void may be represented in a cube as shown in the figure. In which there spheres form the triangular base, the fourth lies at the top and the sphere occupies the tetrahedral void.

Let the length of the side of the cube = a



From right angled triangle ABC, face diagonal

$$AB = \sqrt{AC^2 + BC^2} = \sqrt{a^2 + a^2} = \sqrt{2}a$$

As spheres A and B are actually touching each other, face diagonal AB = 2R

$$\therefore \quad 2R = \sqrt{2}a \quad \text{or} \quad R = \frac{1}{\sqrt{2}}a \quad ...(i)$$

Again from the right angled triangle ABD

$$AD = \sqrt{AB^2 + BD^2} = \sqrt{\left(\sqrt{2}a\right)^2 + a^2} = \sqrt{3}a$$

But as small sphere (void) touches other spheres, evidently body diagonal AD = 2(R + r).

$$\therefore \ 2(R+r) = \sqrt{3}a$$

$$\Rightarrow \ R + r = \frac{\sqrt{3}}{2}a \qquad ...(ii)$$

Dividing equation (ii) by equation (i)

$$\frac{R+r}{R} = \frac{\sqrt{3}/2 \times a}{a/\sqrt{2}} = \frac{\sqrt{3}}{\sqrt{2}}$$

$$\Rightarrow 1 + \frac{r}{R} = \frac{\sqrt{3}}{\sqrt{2}} = 1.225 \quad \Rightarrow \frac{r}{R} = 1.225 - 1 = 0.225$$

$$\Rightarrow r = 0.225\ R$$

RADIUS RATIO IN 1:1 OR AB TYPE STRUCTURE

Radius ratio r^+/r^-	Structural Arrangement	Coordination number	Example
0.225 – 0.414	Tetrahedral	4	CuCl, CuBr, CuI, BaS, HgS
0.414 – 0.732	Octahedron	6	MgO, NaBr, CaS, MnO, KBr, CaO
0.732 – 1	Cubic	8	CsI, CsBr, TlBr, NH_4Br

Illustration 20. *The two ions A^+ and B^- have radii 88 and 200 pm respectively. In the close packed crystal of compound AB, predict the coordination number of A^+.*

Solution: $\frac{r^+}{r^-} = \frac{88}{200} = 0.44$

It lies in the range of 0.414 – 0.732

Hence, the coordination number of $A^+ = 6$

Illustration 21. *Br^- ion forms a close packed structure. If the radius of Br^- ions is 195 pm. Calculate the radius of the cation that just fits into the tetrahedral hole. Can a cation A^+ having a radius of 82 pm be slipped into the octahedral hole of the crystal A^+Br^-?*

Solution:

(i) Radius of the cations just filling into the tetrahedral hole
= Radius of the tetrahedral hole = 0.225×195
= 43.875 pm

(ii) For cation A^+ with radius = 82 pm

Radius ratio $\frac{r^+}{r^-} = \frac{82}{195} = 0.4205$

As it lies in the range 0.414 – 0.732, hence the cation A^+ can be slipped into the octahedral hole of the crystal A^+Br^-.

Illustration 22. *Why is co-ordination number of 12 not found in ionic crystals?*

Solution: Maximum radius ratio in ionic crystals lies in the range 0.732 – 1 which corresponds to a coordination number of 8. Hence coordination number greater than 8 is not possible in ionic crystals.

Illustration 23. *Iron changes its crystal structure from body-centred to cubic close-packed structure when heated to 916°C. Calculate the ratio of the density of the bcc crystal to that of ccp crystal, assuming that the metallic radius of the atom does not change.*

Solution: n the bcc packing, the space occupies is 68% of the total volume available while in ccp, the space occupied is 74%. This means that for the same volume masses of bcc and ccp are in the ratio of 68 : 74. As the volume is same, ratio of density is also same viz 68 : 74

i.e. $\frac{d(bcc)}{d(ccp)} = \frac{68}{74} = 0.919$

Illustration 24. *In BeO (Zinc Blende structure), Mg^{2+} is introduced in available tetrahedral voids. Then ions are removed from a single body diagonal of the unit cell. What will be the molecular formula of the unit cell?*

Solution: In BeO (Zinc Blende structure), alternative 'Td' voids of FCC of O^{2-} ions are occupied by Be^{2+}. So Mg^{2+} can be introduced in four available alternative 'Td'

voids and unit cell formula will be $Be_4Mg_4O_4$. If now ions are removed from a single body diagonal then unit cell formula will be $Be_3Mg_3O_{3.75}$.

Illustration 25. *In a crystal oxide ions are arranged in fcc and A^{+2} ions are at 1/8th of the tetrahedral voids, and ions B^{+3} occupied ½ of the octahedral voids. Calculate the packing fraction of the crystal if O^{-2} of the removed from alternate corner and A^{+2} is being place at 2 of the corners.*

Solution: Since oxide ions are fcc so $4O^{-2}$ / unit cell
A^{+2} are at 1/8th of the tetrahedral so $1A^{+2}$|unitcell
B^{+3} occupies ½ of the octahedral voids $\therefore$ $2B^{+2}$/unit cell
After removing O^{-2} ions

Oxide ion / unit cell = $\frac{4}{8}+3 = 3.5$

A^{+2} ions/ unit cell = $\frac{2}{8}+1 = \frac{10}{8} = 1.25$

B^{+3} ions/unit cell = 2

$$\therefore \text{P.F} = \frac{3.5\times\frac{4}{3}\pi r_-^3 + 1.25\times\frac{4}{3}\pi r_{A^{+2}}^3 + 2\times\frac{4}{3}\pi r_{B^{+3}}^3}{a^3}$$

We know that $a = \frac{4r_-}{\sqrt{2}}$

$\frac{r_{A^{+2}}}{r_-} = 0.225$, $\quad \therefore r_{A^+} = 0.225 r_-$

$\frac{r_{B^{+3}}}{r_-} = 0.414 \therefore r_{B^{+3}} = 0.414 r_-$

Putting all the values
P.F = 0.676

Illustration 26. *A binary solid (A^+B^-) has a rock salt structure. If the edge length is 400 pm, and radius of cation is 75 pm the radius of anion is*

(A) 100 mm *(B) 125 pm*
(C) 250 pm *(D) 325 pm*

Solution: **(B)** Edge = $2(r^+ + r^-)$

$\Rightarrow 400 = 2(75 + r^-)$

$\therefore r^- = 125$ pm

Illustration 27. *In closest packing of atoms*

(A) The size of tetrahedral void is greater than that of the octahedral void.
(B) The size of the tetrahedral void is smaller than that of the octahedral void.
(C) The size of tetrahedral void is equal to that of the octahedral void.
(D) The size of tetrahedral void may be larger or smaller or equal to that of the octahedral void depending upon the size of atoms.

Solution: **(B)** For tetrahedral voids

$\frac{r_+}{r_-} = 0.225$, $r_+ = 0.225\ r_-$ …(i)

Similarly for octahedral voids

$r_+ = 0.414\ r_-$ …(ii)

From equation (i) and (ii) it is clear that size of octahedral void is larger than that of tetrahedral voids.

Exercise 7.

(i) If the radius of Mg^{+2} ion, Cs^{+} ion, O^{2-} ion S^{-2} ion and Cl^{-} ion are 0.65A°, 1.69A°, 1.40A°, 1.84A°, and 1.81A° respectively. Calculate the coordination number of the cation in the crystals of MgS, MgO and CsCl.

(ii) Predict the structure of MgO crystal and coordination number of its cation in which cation and anion radii respectively are to 65 pm and 140 pm.

Exercise 8.

The radius of calcium ion is 94 pm and of an oxide ion is 146 pm. The coordination number of calcium is ……

CALCULATION INVOLVING UNIT CELL DIMENSIONS

From the unit cell dimensions, it is possible to calculate the volume of the unit cell. Knowing the density of the metal. We can calculate the mass of the atoms in the unit cell. The determination of the mass of a single atom gives an accurate determination of Avogadro constant.

Suppose edge of unit cell of a cubic crystal determined by X – Ray diffraction is a, d is density of the solid substance and M is the molar mass, then in case of cubic crystal

Volume of a unit cell = a^3

Mass of the unit cell = no. of atoms in the unit cell $\times$ mass of each atom = Z $\times$m

Here Z = no. of atoms present in one unit cell

m = mass of a single atom

Mass of an atom present in the unit cell = $\frac{m}{N_A}$

$\therefore$ Density d = $\frac{\text{mass of unit cell}}{\text{volume of unit cell}} = \frac{Z.m}{a^3}$

d = $\frac{Z.M}{a^3 \times N_A}$

Note:

Density of the unit cell is same as the density of the substance

Illustration 28. *An element having atomic mass 60 has face centred cubic unit cell. The edge length of the unit cell is 400 pm. Find out the density of the element?*

***Solution*:** Unit cell edge length = 400 pm

= 400×10^{-10} cm

Volume of unit cell = $(400\times10^{-10})^3 = 64\times10^{-24}$ cm^3

Mass of the unit cell = No. of atoms in the unit cell $\times$ mass of each atom

No. of atoms in fcc unit cell = $8\times\frac{1}{8}+6\times\frac{1}{2}=4$

$\therefore$ Mass of unit cell = $\frac{4\times60}{6.023\times10^{23}}$

Density of unit cell =

$\frac{\text{mass of unit cell}}{\text{volume of unit cell}} = \frac{4\times60}{6.023\times10^{23}\times64\times10^{-24}} = 6.2\,g/cm^3$

Illustration 29. *An element has a body centred cubic (bcc) structure with a cell edge of 288 pm. The density of the element is 7.2 g/cm³. How many atoms are present in 208 g of the element?*

Solution: Volume of unit cell = $(288\times10^{-10})^3$ cm^3 = 2.39×10^{-23}cm^3

Volume of 208 g of the element = $\frac{\text{mass}}{\text{volume}} = \frac{208}{7.2} = 28.88\text{cm}^3$

No of unit cells in this volume = $\frac{28.88}{2.39\times10^{-23}} = 12.08\times10^{23}$

Since each bcc unit cell contains 2 atoms

$\therefore$ no of atom in 208 g = $2\times12.08\times10^{23} = 24.16\times10^{23}$ atom

Illustration 30. *A compound formed by elements X & Y, Crystallizes in the cubic structure, where X is at the corners of the cube and Y is at the six face centers. What is the formula of the compound? If side length is 5A°, estimate the density of the solid assuming atomic weight of X and Y as 60 and 90 respectively.*

Solution: From eight corner atoms one atoms (X) contributes to one unit cell.

From six face centres, three atoms (Y) contributes to one unit cell.

So, the formula of the compound is XY_3.

As we know that,

$\rho = \frac{n\times M_m}{N_A\times a^3}$, here n = 1

Molar mass of XY_3

$M_m = 60 + 3\times 90 = 330$ gm

$\rho = \frac{1\times330}{6.023\times10^{23}\times(5\times10^{-8})^3}$ gm/cm^3

a = 5Å = 5×10^{-8} cm

$= \frac{330}{6.023\times10^{23}\times125\times10^{-24}}$ gm/cm^3 **= 4.38 gm / cm^3**

Illustration 31. *Lithium borohydrides, $LiBH_4$, crystallizes in an orthorhombic system with 4 molecules per unit cell. The unit cell dimensions are: a = 6.81 Å, b = 4.43 Å and c = 7.17 Å. Calculate the density of the crystal. Take atomic mass of Li = 7, B = 11 and H = 1 a.m.u.*

Solution: Molar mass of $LiBH_4$ = 7 + 11 + 4 = 22 g mol^{-1}

Mass of the unit cell $= \frac{4\times22\text{ gmol}^{-1}}{6.02\times10^{23}\text{ mol}^{-1}} = 14.62\times10^{--23}$ g

Volume of the unit cell = a × b × c

= (6.81×10^{-8} cm) (4.43×10^{-8} cm) (7.17×10^{-8} cm)

= 21.63×10^{-23} cm^3

$\therefore$ Density of the unit cell $= \frac{\text{Mass}}{\text{Volume}} = \frac{14.62\times10^{-23}\text{ g}}{21.63\times10^{-23}\text{ cm}^3} = 0.676\text{ gcm}^{-3}$

Illustration 32. *A binary solid (A^+B^-) has a rocksalt structure. If the edge length is 400 pm and the radius of cation is 75 pm, the radius of anion is*

(A) 100 pm *(B) 125 pm*
(C) 250 pm *(D) 325 pm*

Solution: **(B)**

Illustration 33. *The vacant space in bcc lattice unit cell is about*
(A) 32% *(B) 10%*
(C) 23% *(D) 46%*

Solution: **(A)**

Illustration 34. *A substance has density of 2 kg dm^{-3} & it crystallizes to fcc lattice with edge-length equal to 700pm, then the molar mass of the substance is*
(A) 74.50gm mol^{-1} *(B) 103.30gm mol^{-1}*
(C) 56.02gm mol^{-1} *(D) 65.36gm mol^{-1}*

Solution: **(B)** $\rho = \dfrac{n \times M_m}{N_A \times a^3}$

$$2 = \frac{4 \times M_m}{6.023 \times 10^{23} \times (7 \times 10^{-8})^3}$$

(since, effective number of atoms in unit cell = 4)
On solving we get M_m = **103.03 gm / mol**

Exercise 9.
(i) Iron has body centred cubic lattice structure. The edge length of the unit cell is found to be 286 pm. What is the radius of an iron atom?
(ii) KF has rock – salt structure
Calculate the value of Avogadro's number. Density of KF = 2.48 g/cm^3
And distance between K^+ & F^- in KF = 268 pm

Exercise 10.
An element crystallises as F.C.C. with density as 5.20 g/cc and edge length of the side of unit cell as 300 pm. Calculate mass of that element which contains 3.01×10^{24} atoms.

STRUCTURE DETERMINATION BY X – RAYS

The German physicist M Von Laue (1879 – 1960) in 1913 suggested the possibility of diffraction of X – Rays by crystals. The reason for this suggestion was that the wavelength of X – Rays was of about the same order as the inter atomic distances in a crystal. Then if these X – Rays was allowed to strike the crystal the rays will penetrate into the crystal and will be scattered by the electrons of the atoms or the ions of the crystal. The rays reflected from different layers of the atoms, due to wave nature will then undergo interference (constructive and destructive) to produce a diffraction pattern just as it happens in case of light passing through a grating containing a large number of closely spaced lines. In other words, crystals should act as a three

dimensional grating for X – Rays. Bragg applied this fact in determining structure and dimensions of crystal.

W.L. Bragg and his father W.H. Bragg determined the cubic structure of NaCl using X – Rays. According to Bragg, a crystal (Composed of series of equally spaced atomic planes) could be employed not only as a transmission grating (as suggested by Laue) but also as a reflection grating. In Bragg's treatment, the X – rays strike the crystal at angle θ, these penetrates into the crystal and are reflected by different parallel layers of particles in the crystal.

A strong reflected (constructive) beam will result only if all the reflected rays are in phase. The reflected waves by different layer planes will be in phase with one another only if the difference in the path length of the waves reflected from the successive planes is equal to an integral number of wavelengths.

It may noted from the fig. that the beams of X – rays which are reflected from deeper layers travel more to reach the detector. Two X – Ray waves in phase are shown to be approaching the crystal. One wave is reflected from the first layer of atoms while the second wave is reflected from the second layer of atoms. The wave reflected from the second layer travels more distance before emerging from the crystal than the first wave. The extra distance travelled is equal to LN + NM. For constructive interference to take place the extra distance travelled by the more penetrating beam must be on integral multiple of the wavelength

Path difference = LN + NM

$= LN = n\lambda$ $(n = 1, 2, 3 \ldots.)$

Since the triangles OLN and OMN are congruent hence LN = NM

So, path difference = 2LN

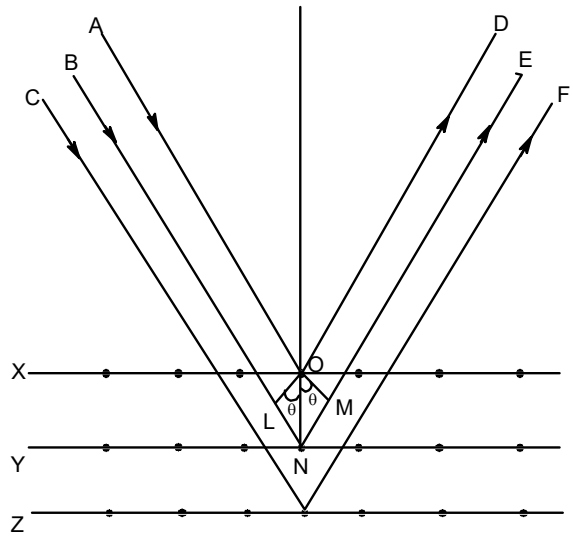
LN = d Sinθ where d is the distance between two planes

So, path difference = 2 d Sinθ

For constructive interference 2 d Sinθ must be equal to $n\lambda$

Or, $\mathbf{n\lambda = 2\ d\ Sin\theta}$

This relation is called Bragg's equation. Distance between two successive planes d can be calculated form this equation. With X – Ray of definite wavelength, reflection at various angles will be observed for a given set of planes separated by a distance'd'. These reflections correspond to n = 1, 2 , 3 and so on and are spoken of as first order, second order, third order and so on the angle θ increases as the intensity of the reflected beams weakens



X – Ray reflection from crystals

Illustration 35. *X – Rays of wavelength 1.54A° strike a crystal and are observed to be deflected at an angel 22.5°. Assuming that n = 1, calculate the spacing between the planes of atom that are responsible for this reflection.*

Solution: Applying Bragg's equation

$n\lambda = 2d \sin\theta$

Giving $n = 1$, $\lambda = 1.54A°$, $\theta = 22.5°$

Using relation $n\lambda = 2 d \sin\theta$

$$d = \frac{1.54}{2\sin 22.5°} = \frac{1.54}{2 \times 0.383} = 2.01A°$$

Illustration 36. *Calculate the angle at which second order reflection will occur in an X- ray spectrometer. When X – ray of wavelength 1.54 A° are diffracted by atoms of a crystal. The interplanar distance is 4.04A°.*

Solution: According to Bragg's equation

$n\lambda = 2 d \sin\theta$

$n = 2$

$\lambda = 1.54A°$; $d = 4.04\ A°$

$$\theta = \sin^{-1}\left(\frac{2 \times 1.54}{2 \times 4.04}\right) \Rightarrow \sin^{-1}\left(\frac{1.54}{4.04}\right)$$

$= \sin^{-1}(0.381)$

$= 22° 24'$

Exercise 11.

(ii) The interplaner distance in a crystal used for X – Ray diffraction is 2A°. The angle of incidence of X – rays is 9.0. When the first order diffraction is observed, calculate the wavelength of the X – rays. (Sin 20° = 0.342)

(ii) The diffraction of a crystal of barium with X – rays of wavelength 2.29A°, gives a first order reflection at 27°8′. What is the distance between the diffracting planes (Sin 27°8′ = 0.4561)

STRUCTURE OF IONIC COMPOUNDS

Simple ionic compounds are of two types i.e. AB and AB_2 type. From the knowledge of close packed structures and the voids developed there in, we can have an idea about the structures of simple ionic compounds.

Among the two ions, constituting the binary compounds, the anions usually constitute the space lattice with hcp or ccp type of arrangements whereas the cations, occupy the interstitial voids

(a) If the anions (B^-) constitute the crystal lattice and all octahedral voids are occupied by cations (A^+), then the formula of the ionic solid is AB.

(b) Similarly, if half of the tetrahedral voids are occupied by cations, then the formula of the solid crystal becomes A^+B^-.

(c) When the anions (B^{-2}) are constituting space lattice and all the tetrahedral voids are occupied by the cations (A^+), then the formula of the solid crystal will be A_2B.

Ionic compounds of the type AB

Ionic compounds of the type AB have three types of crystalline structures. (A) ZnS type (B) NaCl types (C) CsCl types

1. Sodium chloride (Rock salt) type structure

The sodium chloride structure is composed of Na^+ and Cl^- ions. The number of sodium ions is equal to that of Cl^- ions. The radii of Na^+ and Cl^- ions 95 pm and 181 pm giving the radius ratio of 0.524

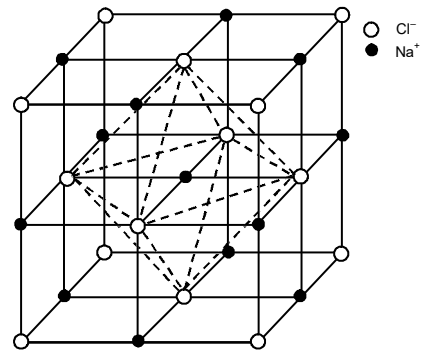
$$\frac{r_{Na^+}}{r_{Cl^-}} = \frac{95}{181} = 0.524$$

The radius ratio of 0.524 for NaCl suggest an octahedral void. Thus the salient features of this structure are as follows:

(i) Chloride ions (In a typical unit cell) are arranged in cubic close packing (ccp). In this arrangement, Cl^- ions are present at the corners and at the centre of each face of the cube. This arrangement is also regarded as face centred cubic arrangement (fcc).

(ii) The sodium ions are present in all the octahedral holes.

(iii) Since, the number of octahedral holes in ccp structure is equal to the number of anions, every octahedral hole is occupied by Na^+ ions. So that the formula of sodium chloride is NaCl i.e. stoichiometry of NaCl is 1:1.



Unit cell structure of NaCl

(iv) Since there are six octahedral holes around each chloride ions, each Cl^- ion is surrounded by 6 Na^+ ions. Similarly each Na^+ ion is surrounded by 6 Cl^- ions. Therefore, the coordination number of Cl^- as well as of Na^+ ions is six. This is called 6:6 coordination.

(v) It should be noted that Na^+ ions to exactly fit the octahedral holes, the radius ratio $\frac{r_{Na^+}}{r_{Cl^-}}$ should be equal to 0.414. However, the actual radius ratio $\left(\frac{r_{Na^+}}{r_{Cl^-}} = 0.524\right)$ exceeds this value. Therefore to accommodate large Na^+ ions, the Cl^- ions move apart slightly i.e. they do not touch each other and form an expanded face centred lattice.

(vi) The unit cell of sodium chloride has 4 sodium and 4 chloride ions as calculated below
No of sodium ions = 12 (at edge centres) $\times$1/4 + 1 (at body centre)$\times$1= 4
No of chloride ions = 8(at corner)$\times$1/8+6(at face centres)$\times$1/2 = 4

Thus, the number of NaCl units per unit cell is 4.

(vii) The edge length of the unit cell of NaCl type of crystal is 2(r+R) (r = radii of Na^+ ion)

i.e. (R = radii of Cl^- ion)

$$a = 2\left(r_{Na^+} + r_{Cl^-}\right)$$

Thus, the distance between Na^+ and Cl^- ions = a/2

Most of the halides of alkali metals, oxides and sulphides of alkaline earth metals have this type of structures. Some of the common examples are NaI, KCl, RbI, RbF, NH_4Cl, NH_4Br, AgCl, AgBr and AgI.

Ferrous oxide also has sodium chloride, types structure in which O^{-2} ions are arranged in ccp and Fe^{+2} ions occupy octahedral ions. However, this oxide is always non – stoichiometric and has the composition $Fe_{0.95}O$. It can be explained on the assumption that some of the Fe^{+2} ion are replaced by 2/3rd as many Fe^{+3} ions in the octahedral voids.

2. Zinc blende (ZnS) type structures (sphelerite)

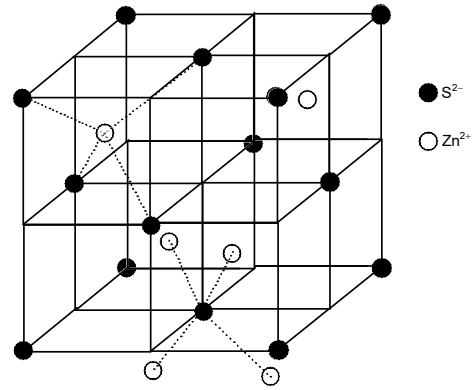
The zinc sulphide crystals are composed of equal number of Zn^{+2} and S^{2-} ions. The radii of the two ions (Zn^{+2} = 74 pm and S^{-2} = 184 pm) led to the radius $\left(r^+/r^-\right)$ as 0.40 which suggests a tetrahedral arrangement.

$$\frac{r_{Zn^{+2}}}{r_{S^{-2}}} = \frac{74}{184} = 0.40$$

The salient features of this structure are as follows

(i) The Zinc ions are arranged in ccp arrangement, i.e. sulphide ions are present at the corners and the centres of each face of the cube

(ii) Zinc ions occupy tetrahedral hole. Only half of the tetrahedral holes are occupied by Zn^{+2} so that the formula of the zinc sulphide is ZnS i.e. the stoichiometry of the compound is 1:1 (Only alternate tetrahedral holes are occupied by Zn^{+2})



Zinc blende structure zinc sulphide

(iii) Since the void is tetrahedral, each zinc ion is surrounded by four sulphide ions and each sulphide ion is surrounded tetrahedrally by four zinc ions. Thus zinc sulphide has 4:4 Co – ordination.

(iv) For exact fitting of Zn^{+2} in the tetrahedral holes, formed by close packing of S^{-2} ions, the ratio Zn^{+2}/S^{-2} should be 0.225. Actually this ratio is slightly large (0.40)

(v) There are four Zn^{+2} ions and four S^{-2} ions per unit cell as calculated below:

No. of S^{-2} ions = 8(at corners)×1/8 + 6(at face centres)×1/2 = 4

No. of Zn^{+2} ions = 4(within the body)×1 = 4

Thus, the number of ZnS units per unit cell is equal to 4. Some more examples of ionic solids having Zinc blende structures are CuC, CuBr, CuI, AgI, beryllium sulphide.

Illustration 37. *If silver iodide crystallizes in a zinc blende structure with I^- ions forming the lattice then calculate fraction of the tetrahedral voids occupied by Ag^+ ions.*

Solution: In AgI, if there are nI^- ions, there will be nAg^+ ions. As I^- ions form the lattice, number of tetrahedral voids = 2n. As there are nAg^+ ions to occupy these voids, therefore fraction of tetrahedral voids occupied by Ag^+ ions = n/2n = ½ = 50%.

3. The Wurtzite structures

It is an alternate form in which ZnS occurs in nature. The main features of this structure are

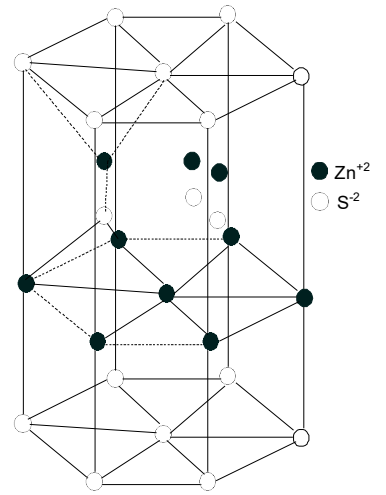
(i) Sulphide ions have HCP arrangement and zinc ions occupy tetrahedral voids.

(ii) Only half the alternate tetrahedral voids are occupied by Zn^{+2} ions.

(iii) Coordinate no. of Zn^{+2} ions as well as S^{-2} ions is 4. Thus, this structure has 4 : 4 coordination.

(iv) No. of Zn^{+2} ions per unit cell

= 4(within the unit cell) $\times 1$ + 6(at edge centres) $\times \frac{1}{3}$ = 6



A unit cell representation of wurtzite structure

No. of S^{-2} ions per unit cell

$$= 12(\text{at corners}) \times \frac{1}{6} + 2(\text{at face centres}) \times \frac{1}{2} + 3(\text{within the unit cell}) \times 1 = 6$$

Thus, there are 6 formula units per unit cell.

Illustration 38. *The co–ordination number of a metal crystallising in hcp structure is*

(A) 12 *(B) 10*

(C) 8 *(D) 6*

Solution: **(A)**

4. Caesium chloride (CsCl) structure

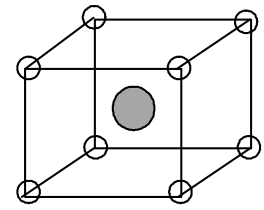
The caesium chloride crystal is composed of equal number of caesium (Cs^+) and Chloride Cl^- ions. The radii of two ions (Cs^+ = 169 pm and Cl^- = 181 pm) led to radius ratio of r_{Cs^+} to r_{Cl^-} as 0.93 which suggest a body centred cubic structure having a cubic hole

$$\frac{r_{Cs^+}}{r_{Cl^-}} = \frac{169}{181} = 0.93$$

The salient features of this structure are as follows:

(i) The chloride ion form the simple cubic arrangement and the caesium ions occupy the cubic interstitial holes. In other words Cl^- ions are at the corners of a cube whereas Cs^+ ion is at the centre of the cube or vice versa

(ii) Each Cs^+ ion is surrounded by 8 Cl^- ions and each Cl^- ion in surrounded by 8 Cs^+ ions. Thus the Co – ordination number of each ion is eight.



Caesium halide structure

(iii) For exact fitting of Cs^+ ions in the cubic voids the ratio $\frac{r_{Cs^+}}{r_{Cl^-}}$ should be equal to 0.732. However, actually the ratio is slightly larger (0.93). Therefore packing of Cl^- ions slightly open up to accommodate Cs^+ ions.

(iv) The unit cell of caesium chloride has one Cs^+ ion and one Cl^- ion as calculated below
No. of Cl^- ion 8(at corners) $\times 1/8 = 1$
No. of Cs^+ ion = 1(at body centre)$\times 1=1$
Thus, number of CsCl units per unit cell is 1

(v) Relation between radius of cation and anion and edge length of the cube, $r_{Cs} + r_{Cl} = \frac{a\sqrt{3}}{2}$

Other common examples of this type of structure are CsBr, CsI, TlCl, TlBr

Higher coordination number in CsCl(8:8) suggest that the caesium chloride lattice is more stable than the sodium chloride lattice in which Co – ordination number is 6:6. Actually the caesium chloride lattice is found to be 1% more stable than the sodium chloride lattice. Then the question arises why NaCl and other similar compounds do not have CsCl type lattice – This is due to their smaller radius ratio. Any attempt to pack 8 anions around the relatively small cation (Li^+, Na^+, K^+, Rb^+) will produce a state in which negative ions will touch each other, sooner they approach a positive ion. This causes unstability to the lattice.

Effect of temperature on crystal structure

Increase of temperature decreases the coordination of number, e.g. upon heating to 760 K, the CsCl type crystal structure having coordination 8:8 changed to NaCl type crystal structures having coordination 6:6.

$$\underset{(8:8\text{ coordination})}{\text{CsCl type crystal}} \xrightarrow{\text{high temp}} \underset{(6:6\,\text{Co-ordination})}{\text{NaCl type crystal}}$$

Effect of pressure on crystal structure

Increase of pressure increases the Co – ordination number during crystallization e.g. by applying pressure, the NaCl type crystal structure having 6:6 coordination number changes to CsCl type crystal having coordination number 8:8

$$\underset{(8:8\text{ coordination})}{\text{NaCl type crystal}} \xrightarrow{\text{high pressure}} \underset{(6:6\,\text{Co-ordination})}{\text{CsCl type crystal}}$$

Illustration 39. *Out of NaCl and CsCl, which one is more stable and why?*

Solution: CsCl is more stable than NaCl. This is because higher the coordination number, greater are the forces of attraction between the cations and the anions in the close-packed arrangement. As CsCl has co-ordination number of 8 : 8 while NaCl has a coordination number of 6 : 6, therefore CsCl is more stable.

Illustration 40. *Compare the structure of zinc blend (ZnS) with that of diamond.*

Solution: The structure of zinc blend (ZnS) is similar to that of diamond. In diamond each carbon atom is linked to four other atoms tetrahedrally. Similarly, in zinc blende, each Zn^+ ion is surrounded by $4s^{2-}$ ions tetrahedrally and each S^{2-} ion is surrounded by $4Zn^{2+}$ ions tetrahedrally. Thus if all Zn^{2+} ions and S^{2-} ions in zinc blende are replaced by carbon atoms, if gives rise to structure of diamond.

Illustration 41. Match **LIST – X** with **LIST – Y**

	LIST – X	LIST – Y
(A)	Zinc blende structure	(A) oxide ions in hcp and 2/3rd of octahedral voids occupied by trivalent cation.
(B)	Anti–fluorite structure	(B) anion in FCC arrangement and alternate tetrahedral voids occupied by bivalent cation.
(C)	Corundum structure	(C) Oxide ions in CCP, bivalent cations in corners and one tetravalent cation in the octahedral void created by oxide ions.
(D)	Pervoskite structure	(D) Anions in FCC and cations in all tetrahedral voids

(A) AB, BC, CD, DA (B) AB, BD, CA, DC
(C) AD, BA, CB, DC (D) AC, BD, CB, DA

Solution: **(B)**

Exercise 12.
MgO has a structure of NaCl and TlCl has the structure of CsCl. What are the coordination numbers of ions in each?

IONIC COMPOUND OF THE TYPE AB_2

Calcium fluoride (Fluorite) structure

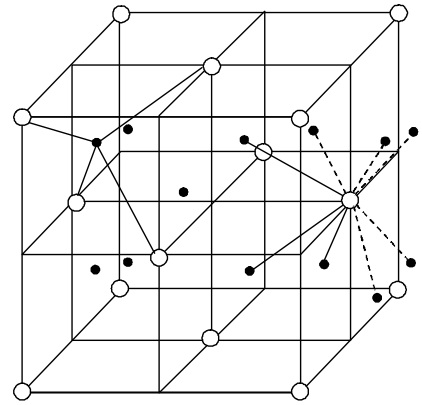
The salient features of fluorite structure are

(i) The Ca^{+2} ions are arranged in ccp arrangement, i.e. these ions occupy all the corners and the centres of each face of the cube

(ii) The F^- ions occupy all the tetrahedral holes.

(iii) Since there are two tetrahedral holes for each Ca^{+2} ion and F^- ions occupy all the tetrahedral holes, there will be two F^- ions for each Ca^{+2} ions, thus the stoichiometry of the compound is 1:2.

(iv) Each Ca^{+2} ion is surrounded by $8F^-$ ions and each F^- ions is surrounded by $4Ca^{+2}$ ions. The Coordination number of Ca^{+2} ion is eight and that of F^- ion is four, this is called 8:4 Coordination.



Unit cell representation of CaF_2 structure

(v) Each unit cell has 4 calcium ions and 8 fluoride ions as explained below
No. of Ca^{+2} ions = 8(at corners)×1/8 + 6 (at face centres)×1/2
No. of F ions = 8 (within the body)×1 = 8
Thus the number of CaF_2 units per unit cell is 4.
Other examples of structure are SrF_2, $BaCl_2$, BaF_2, PbF_2, CdF_2, HgF_2, CuF_2, $SrCl_2$, etc.

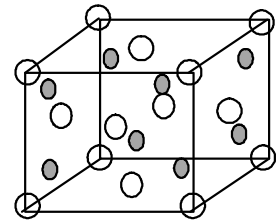
Ionic compound of A_2B type

The compound having A_2B formula are compounds having anti fluorite structure
Anti fluorite structure is having arrangement of cations and anions opposite to the fluorite structure Li_2O has an anti fluorite structure.

(i) In the crystal structure of Li_2O, the O^{-2} ions constitute a cubic close packed lattice (fcc structure) and the Li^+ ions occupy all the tetrahedral voids

(ii) Each oxide ion, O^{-2} ion is in contact with 8 Li^+ ions and each Li^+ ions having contact with 4 oxide ion. Therefore, Li_2O has 4:8 coordination

Examples – Na_2O, K_2O, K_2S, Na_2S, Rb_2O, Rb_2S



Anti-fluorite structure

Normal spinel structure

Spinel is a mineral $MgAl_2O_4$. In it oxide ions are arranged in ccp with Mg^{+2} ions occupying tetrahedral voids and Al^{+3} ions in a set of octahedral voids. Many ferrites (such as $ZnFe_2O_4$) also possess spinel structure. These are very important magnetic materials and are used in telephone and memory loops in computers.

Structure of Fe_3O_4 (Magnetite)

In Fe_3O_4, Fe^{+2} and Fe^{+3} ions are present in the ratio 2:1. it may be considered as having composition $FeO.Fe_2O_3$. In Fe_3O_4 Oxide arranged in ccp. Fe^{+2} ions occupy octahedral voids while Fe^{+3} ions are equally distributed between octahedral and tetrahedral voids

$MgFe_2O_4$ also has structure similar to magnetite. In this Mg^{+2} ions are present in place of Fe^{+2} ion in Fe_3O_4. Magnetite has inverse spinet structure.

SUMMARY OF VARIOUS STRUCTURES OF IONIC SOLIDS

Crystal structure		Brief description	Coordination number	No. of atoms per unit cell	Examples
1.	Rock salt (NaCl – type)	Cl^- ions in ccp Na^+ ions occupy all octahedral voids	$Na^+ = 6$ $Cl^- = 6$	4	Li, Na, KI, and Rb halides NH_4Cl, NH_4Br, NH_4I, AgF, AgCl, AgBr, MgO, CaO, TiO, FeO, NiO
2.	Zinc blende (ZnS – types)	S^{-2} ions in ccp Zn^{+2} ions occupy alternate tetrahedral voids	$Zn^{+2} = 4$ $S^{-2} = 4$	4	ZnS, BeS, CuCl, CuBr, CuI, AgI, HgS
3.	Wurtzite (ZnS – type)	S^{-2} ions in hcp Zn^{+2} ion occupy alternate tetrahedral voids	$Zn^{+2} = 4$ $S^{-2} = 4$	4	ZnS, ZnO, CdS, BeO
4.	Caesium chloride (CsCl type)	Cl^- ions in bcc Cs^+ ions in the body of cube	$Cs^{+2} = 8$ $Cl^- = 4$	1	CsCl, CsBr, CsI, CsCN,CaS
5.	Fluorite (CaF_2 type)	Ca^{+2} ions in ccp, F^- ions occupy all tetrahedral voids	$Ca^{+2} = 8$ $F^- = 4$	4	CaF_2, SrF_2, BaF_2, $BaCl_2$, $SrCl_2$, CdF_2, HgF_2
6.	Anti fluorite	O^{-2} ions in ccp,	$Li^+ = 4$	4	K_2O, Na_2O, K_2S, Na_2S

	(Li_2O – type)	Li^+ ions occupy all tetrahedral sites	O^{-2} = 8		

Illustration 42. *Compute the percentage void space per unit volume of unit cell in zinc-fluoride structure.*

Solution: Since anions occupy fcc positions and half of the tetrahedral holes are occupied by cations.

Since there are four anions and 8 tetrahedral holes per unit cell, the fraction of volume occupied by spheres/unit volume of the unit cell is

$$= \frac{4\times\left(\frac{4}{3}\pi r_a^3\right)+\frac{1}{2}\times 8\times\left(\frac{4}{3}\pi r_c^3\right)}{16\sqrt{2}r_a^3} = \frac{\pi}{3\sqrt{2}}\left\{1+\left(\frac{r_c}{r_a}\right)^3\right\}$$

$\because$ for tetrahedral holes,

$$\frac{r_c}{r_a} = 0.225 = \frac{\pi}{3\sqrt{2}}\ \{1 + (0.225)^3\} = 0.7496$$

$\therefore$ Void volume = 1 – 0.7496 = 0.2504/unit volume of unit cell

% void space = 25.04%

Illustration 43. *Select the correct statements:-*

(A) For CsCl unit cell (edge-length = a), $r_c + r_a = \frac{\sqrt{3}}{2}a$

(B) For NaCl unit cell (edge-length = ℓ*),* $r_c + r_a = \frac{\ell}{2}$

(C) The void space in a b.c.c. unit cell is 0.68

(D) The void space % in a face-centered unit cell is 26%

Solution: In bcc structure are $r_+ + r_- = \frac{\sqrt{3}}{2}a$

Hence, for CsCl, $r_C + r_a = \frac{\sqrt{3}}{2}\mathbf{a}$

$\therefore$**(A)**

Since, NaCl crystallises in fcc structure

$\therefore$ $2r_C + 2r_a$ = edge length of the unit cell

Hence, $r_C + r_a = \frac{\boldsymbol{l}}{\mathbf{2}}$

$\therefore$ **(B)**

Since packing fraction of a bcc unit cell is 0.68

$\therefore$ void space = 1–0.68 = 0.32

$\therefore$**(C)**

In fcc unit cell PF = 74%

$\therefore$ VF = 100–74 = 26%

$\therefore$**(D)**

Illustration 44. Addition of $CdCl_2$ to AgCl yields solid solutions, where the divalent cations Cd^{+2} occupy the Ag^+ sites which one of the following statements is true

(A) The no. of cationic vacancies is one half of the no. of that of divalent ions added.

(B) The no. of cationic vacancies is one half of the no. of that of divalent ions added.

(C) The no. of anionic vacancies is equal in no. to that of divalent ions added.

(D) No cationic or anionic vacancies are produced.

Solution: To maintain the electrical neutrality, the no. of cationic vacancies is equal to the no. of divalent ions added.

Illustration 45. *CsCl has bcc structure with* Cs^+ *at the centre and* Cl^- *ion at each corner. If* r_{Cs^+} *= 1.69Å and* r_{Cl^-} *= 1.81Å, what is the edge length "a" of the cube?*

(A) 3.50Å *(B) 3.80Å*

(C) 4.04Å *(D) 4.50Å*

Solution: **(C)** Assuming the closest approach between Cs^+ and Cl^- ions, the internuclear separation is one–half of the cubic diagonal i.e.

$$.69 + 1.81 = 3.50 = \frac{a\sqrt{3}}{2}$$

$$\therefore a = \frac{2 \times 3.5}{\sqrt{3}} = 4.04\text{Å}$$

IMPERFECTIONS IN SOLIDS: DEFECTS IN CRYSTALS

Atomic imperfections / point defects:

When deviations exist from the regular (or periodic) arrangements around an atom or a group of atoms in a crystalline substance, the defects are called point defects.

Type of point defects – point defects in a crystal may be classified into three types

(a) Stoichiometric defects

(b) Non – stoichiometry defects

(c) Impurity defects

Stoichiometry defect

The compounds in which the number of cation and anions are exactly in the same ratio as represented by their chemical formula are called stoichiometric compounds. The defects that do not disturb the ratio of cations and anions are called stoichiometric defect.

These are of two types:

1. Schottky defect

If in an ionic crystal of the type $A^+ B^-$, equal number of cations and anions are missing from their lattice. It is called Schottky defect.

This type of defect is shown by highly ionic compounds which have

(i) High Co – ordination number and

(ii) Small difference in the sizes of cations and anions

A few examples of ionic compounds exhibiting Schottky defect are NaCl, KCl, KBr and CsCl.

CONSEQUENCES OF SCHOTTKY DEFECT

(a) As the number of ions decreases as a result of this defect, the mass decreases whereas the volume remains the same. Hence density of the solid decreases
(b) The crystal begins to conduct electricity to a small extent by ionic mechanism
(c) The presence of too many voids lowers lattice energy and the stability of the crystal

2. Frenkel defect:

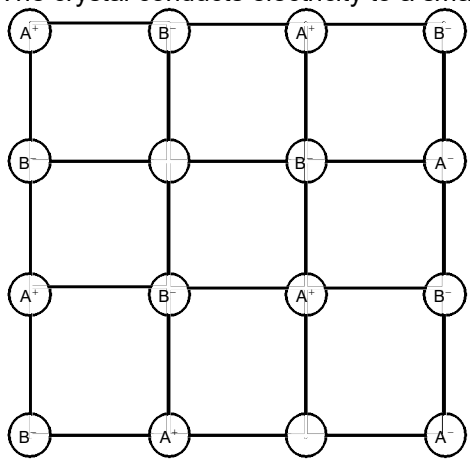
If an ion is missing from its correct lattice sites (causing a vacancy or a hole) and occupies an interstitial site, electrical neutrality as well as stoichiometry of the compounds are maintained. This type of defect is called **Frenkel** defect. Since cations are usually smaller it is more common to find the cations occupying interstitial sites.

This type of defect is present in ionic compounds which have

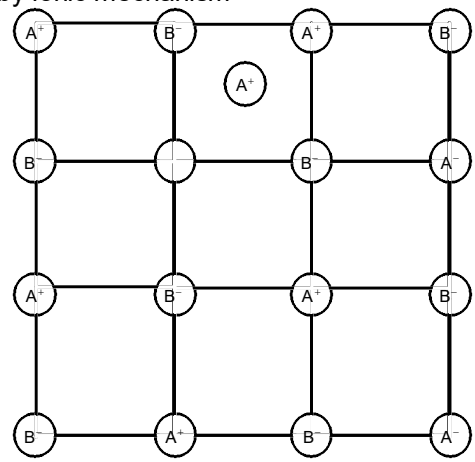
(i) Low co ordinations number
(ii) Larger difference in size of cation and anions
(iii) Compounds having highly polarising cation and easily polarisable anion. A few examples of ionic compounds exhibiting this defect are AgCl, AgBr, AgI, ZnS etc.

Consequences of Frenkel defect

(a) As no ions are missing from the crystal lattice as a whole, therefore density of the solid remains the same
(b) The closeness of like charges tends to increases the dielectric constant of the crystal
(c) The crystal conducts electricity to a small extent by ionic mechanism



Schottkey defect



Frenkel defect

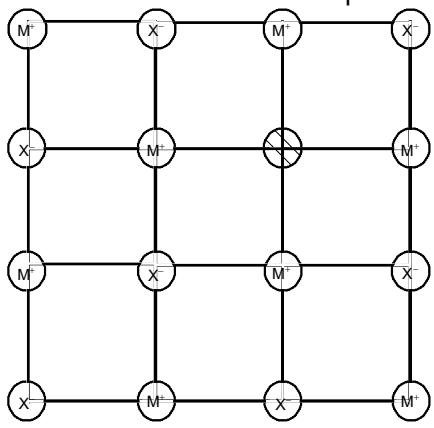
NON – STOICHIOMETRIC DEFECTS

If as a result of imperfection, the ratio of number of cation to anion becomes different from that indicated by the ideal chemical formula, the defects are called non – stoichiometric defects. These defects arise either due to excess of metal atoms or non metal atom or presence of impurities / foreign particle.

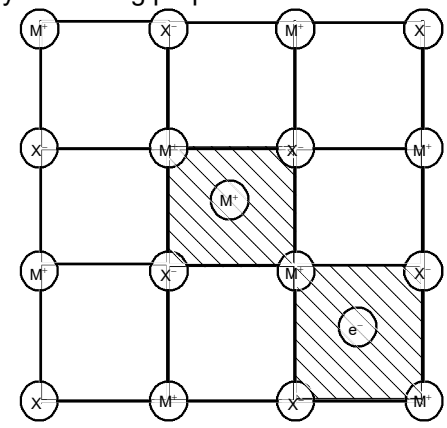
(a) **Metal excess defects due to anion vacancies**: -

A compound may have excess metal ion if a negative ion is absent from its lattice site leaving a hole which is occupied by electron to maintain neutrality.

The holes that are occupied by electrons are called 'F' centres (or colour centres) and are responsible for the colour of the compound and many interesting properties.



Metal excess – due to anion vacancies



Metal excess defect – due to interstitial cation

(ii) **Metal excess defects due to interstitial cations**

Metal excess may also be caused by an extra cation (positive ion) present in an interstitial site. Electrical neutrality is maintained by presence of an electron in another interstitial site. This defect is similar to Frenkel defect and is found in crystals having Frenkel defects.

(iii) **Metal deficiency due to cation vacancies**

The non-stoichiometric compounds may have metal deficiency due to the absence of a metal from its lattice site. The charge is balanced by an adjacent ion having higher positive charge. This type of defects are generally shown by compounds of transition elements.

(c) **Point defects due to the presence of foreign atoms**

These defects arise when foreign atoms are present at the lattice site (in place of host atoms) or at the vacant interstitial sites. In the former case, we get substitutional solid solutions. The formation of former depends upon the electronic structure of impurity while that of later on the size of impurity.

Illustration 46. *Titanium monoxide has a rock–salt structure. X–ray diffraction data show that the length of one edge of the cubic unit cell for TiO with a 1:1 ratio of Ti to O is 4.18Å, and the density as determined by volume and mass measurements is 4.92 g cm^{-3}. Do the data indicate that defects are present? If so, are they vacancy or interstitial defects? [Ti = 47.88 u].*

Solution: The presence of vacancies (Schottky defects) at the Ti and O sites should be reflected in a lower measured density than that calculated from the size of the unit cell and the assumption that every Ti and O site is occupied. Interstitial (Frenkel) defects would give little of any difference between the measured and theoretical densities. There are four formula units per unit all so theoretical density is

$$d = \frac{4(47.88 + 16)}{6.023 \times 10^{23}(4.18 \times 10^{23})} = 4.81 \text{g cm}^{-3}$$

This is significantly greater than the measured density. The crystal must, therefore, contain numerous vacancies. Because the overall composition of the solid is TiO, there must be equal number of vacancies on cation and anion sites.

Illustration 47. *Find out the ratio of the mole-fraction of the Frenkel's defect in NaCl crystal at 4000K temperature. The amount of energy needed to form Frenkel's defects and Schottky defects are respectively 2 eV and 4 eV.*
Given that 1ev = 1.6 × 10^{-19}V
And k = 1.23 × 10^{-23}

Solution: Let the Frenkel defect be n in the ionic crystal of N ions within interstitial space.

Since, $n = \sqrt{N \times Ni}\ \ e^{-\frac{E}{2kT}}$

Since for this crystal

Ni = 2N

$\therefore n = \sqrt{2} \times N \times e^{-\frac{E}{kT}}$

$\therefore \frac{n}{N} = \sqrt{2} e^{-\frac{2\times1.6\times10^{-19}}{2\times1.38\times10^{-23}\times4000}} = \sqrt{2} \times e^{-2.8985}$

$x_1 = \frac{n}{N} = 7.79 \times 10^{-2}$

Now for Schottky defect

Let n defects be present in N-ions of the crystal

$\therefore$ mole fraction $= \frac{n}{N} = e^{-\frac{E}{2kT}}$

$n = e^{-\frac{4\times1.6\times10^{-19}}{2\times1.38\times10^{-23}\times4000}} = e^{-5.797}$

$x_2 = 3.03 \times 10^{-3}$

So, the ratio of mole-fraction of Frenkel's and Schottky defects are

$$\frac{x_1}{x_2} = \frac{7.79\times10^{-2}}{3.03\times10^{-3}} = \frac{25.71}{1}$$

Illustration 48. *Titanium crystallizes in a face centered cubic lattice. It reacts with C or H interstitially, by allowing atoms of these elements to occupy holes in the host lattice. Hydrogen occupies tetrahedral holes, but carbon occupies octahedral holes.*

(a) Predict the formula of titanium hydride and titanium carbide formed by saturating the titanium lattice with either "foreign" element.

(b) What is the maximum ratio of "foreign" atom radius to host atom radius that can be tolerated in a tetrahedral hole without causing a strain in the host lattice?

Solution:

(a) There are 4 – atoms per unit cell and 8-tetrahedral sites per unit cell.
So, the ratio of atoms to tetrahedral sites = 1:2
So, the formula would be **TiH_2**
For carbide
Since carbon occupies octahedral holes
So, ratio of octahedral hole to atom = 1:1
The formula of carbide is **TiC**

(b) Since for tetrahedral hole
The limiting ratios radius
$\frac{r^+}{r^-}$ = 0.225 – 0.414

here r^+ = foreign atom radius

r^- = host atom radius

Without causing a strain in the host lattice i.e., $\frac{r^+}{r^-}$ = 0.225 (minimum value)

PROPERTIES OF SOLIDS

The three main properties of solids which depend upon their structure

(i) Electrical properties

(ii) Magnetic properties

(iii) Dielectric properties

ELECTRICAL PROPERTIES

Electrical conductivity of solids may arise through the motion of electrons and positive holes (electronic conductivity) or through the motions of ions (ionic conductivity). The conduction through electrons is called n-type conduction and through positive holes is called p – types conduction. Electrical conductivity of metal is due to motion of electrons and it increases with the number of electrons available to participate in the conduction process. Pure ionic solids where conduction can take place only through motion of ions are insulators. However, the presence of defects in the crystal structure increases their conductivity.

On the basis of electrical conductivity the solids can be classified into three types

(a) **Metal** (conductors)**:** They allow the maximum portion of the applied electric field to flow through them and have conductivities in order of $10^6 – 10^8$ ohm^{-1}.

(b) **Insulators:** They have low conductivities i.e. they do not practically allow the electric circuit to flow through them. The electrical conductivity is in order $10^{-10} – 10^{-20}$ ohm^{-1} m^{-1}

(c) **Semi conductors:** The solids with intermediate conductivities at the room temperature. Semi conductors allow a portion of electric current to flow through them.
Actually semi conductors are those solids which are perfect insulators at absolute zero, but conduct electric current at room temperature.

(1) **Intrinsic semi conductors** (semi-conductors due to thermal defects)

At zero Kelvin pure substance silicon and germanium act as insulators because electrons fixed in covalent bonds are not available for conduction. However at higher temperature some of the covalent bonds are broken and the electrons so released become free to move in the crystal and thus conduct electric current. This type of conduction is known as intrinsic conduction as it can be introduced in the crystal without adding an external substance.

(2) **Extrinsic semi conductors:** (semi conductors due to impurity defects)

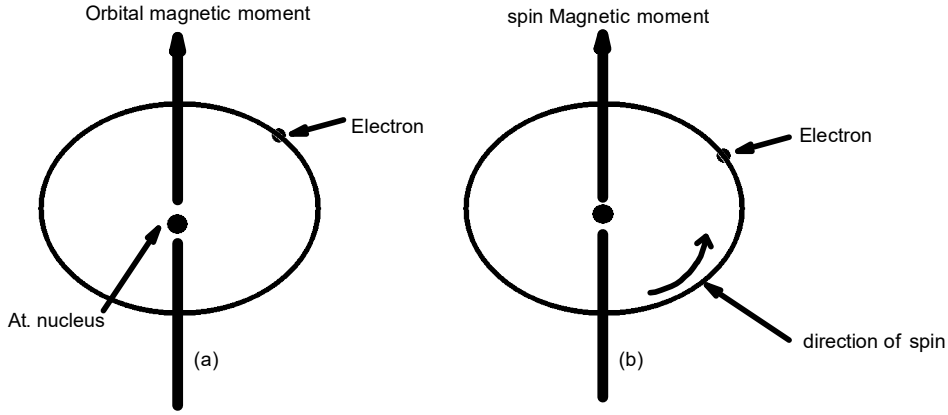
The conductivity of pure silicon and germanium is very low at room temperature. The conductivity of silicon and germanium can be increased by doping with impurities producing n-type semiconductors or p – type semi conductors

MAGNETIC PROPERTIES

The magnetic properties of different materials are studies in terms of their magnetic moments which arise due to the orbital motion and spinning motion of the electron. As electron is charged particle, the circular motion of the electric charge causes the electron to act as a tiny electro

magnet. The magnetic moment of the magnetic field generated due to orbital motion of the electron is along the axis of rotation. The electron also possesses magnetic moment due to the spin which is directed along the spin axis.

Thus, magnetic moment of the electron is due to travelling in closed path (orbital motion) about the nucleus and spinning on its axis. For each electron spin magnetic moment is $\pm\mu B$. Where μ_B, Bohr Magneton is the fundamental unit of magnetic moment and is equal to 9.27×10^{-24} em^2. The magnetic moment due to orbital motion is equal to $M_\ell\,\mu_B$ where M_ℓ is the magnetic quantum number of the electron.



As magnetic moment is a vector quantity, the net magnetic moment of an electron may be represented by an arrow. Thus a material may be considered to contain a number of magnetic dipoles (similar to a bar magnet with north and south poles). Due to the magnetic moment of the electrons different substances behave differently towards the external applied magnetic field. Based on the behaviour in the external magnetic field, the substances are divided into different categories as explained below.

(i) **Diamagnetic substance:**

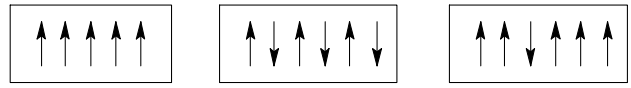
Substances which are weakly repelled by the external magnetic fields are called diamagnetic field e.g. TiO_2, NaCl, benzene etc. Diamagnetic substances have all their electrons paired.

(ii) **Paramagnetic substances:**

Substances which are weakly attracted by magnetic field are called paramagnetic substances. These substance have permanent magnetic dipoles due to the presence of some species (atoms, ions or molecules) with unpaired electron. The paramagnetic substances lose their magnetism in the absence of magnetic field. For e.g. TiO, VO_2 and CuO, O_2, Cu^{+2}, Fe^{+3} etc.

(iii) **Ferromagnetic substances:**

Substances which show permanent magnetism even in the absence of the magnetic field are called Ferromagnetic substances. e.g. Fe Ni. CO, CrO_2 show Ferromagnetism. Such substances remain permanently magnetised, once they have been magnetised. This type of magnetism arises due to spontaneous alignment of magnetic moment due to unpaired electrons in the same direction.



(iv) **Anti Ferromagnetic substance**

Substances which are expected to possess paramagnetism or Ferromagnetism on the basis of unpaired electron but actually they posses zero net magnetic moment are called anti Ferromagnetic substances e.g. MnO, Mn_2O_3, MnO_2.
Anit Ferromagnetism is due to presence of equal number of magnetic moments in the opposite direction.

(v) **Ferrimagnetic substances**

Substance which are expected to posses large magnetism on the basis of the unpaired electrons but actually have small net magnetic moments are called Ferrimagnetic substances e.g. Fe_3O_3

(3) **Dielectric properties**

A dielectric substance is, in which an electric field gives rise to no net flow of electric charge. This is due to the reason that electrons in a dielectric substances are tightly held by individual atoms. However when electric field is applied. Polarization takes place because nuclei are attracted to one side and the electron cloud to the other side. In addition to these dipoles, there may also be permanent dipoles in the crystal.
The alignment of these dipoles may be in compensatory way i.e. the net dipole moment is zero or noncompensatory way i.e. has a net dipole moment. The net dipole moment leads to certain characteristic properties to solids.

(a) **Piezoelectricity (or pressure electricity)**

When mechanical stress is applied on crystals so as to deform them, electricity is produced due to displacement of ions. The electricity thus produced is called piezoelectricity and the crystals are called piezoelectric crystals. Conversely, if electric field is applied to such crystals, atomic displacement takes place resulting into mechanical strain. This is sometimes called **Inverse piezoelectric effect.**
The crystals are used as pick – ups in record players where they produce electrical signals by application of pressure. Examples of piezoelectric crystals include titanates of barium and lead, lead zirconate ($PbZrO_3$), ammonium dihydrogen phosphate ($NH_4H_2PO_4$) and quartz. They are also used in microphones, ultrasonic generators and sonar detectors.

(b) **Pyroelectricity:**

Some piezoelectric crystals when heated produce a small electric current. The electricity thus produced is called pyroelectricity.

(c) **Ferroelectricity:**

In some of the piezoelectric crystals, the dipoles are permanently polarized even in the absence of the electric field. However on applying electric field, the direction of polarization changes e.g. Barium titanate ($BaTiO_3$) sodium potassium tartarate (Rochelle salt) and potassium dihydrogen phosphate (KH_2PO_4). All ferroelectric solids are piezoelectric but the reverse is not true.

(d) **Anti Ferroelectricity:**

In some crystals, the dipoles align themselves in such a way, that alternately, they point up and down so that the crystal does not posses any net dipole moment. Such crystal are said to be anti Ferroelectric e.g. Lead zirconate ($PbZrO_3$)

SUPER CONDUCTIVITY

A substance is said to be superconducting when it offers no resistance to the flow of electricity. Electrical resistance decreases with decreases in temperature and becomes almost zero near the absolute zero. The phenomenon was first discovered by ***Kammerlingh Onnes*** in 1913 when he found that mercury becomes superconducting at 4 K. The temperature at which a substance starts behaving as super conductor is called ***transition temperature***. Most metals have transition temperatures between 2K -5K. Certain organic compounds also becomes superconducting below 5K. Such low temperature can be attained only with liquid helium which is very expensive.

Certain alloys of niobium have been found to be superconducting at temperature as high as 23 K. Since 1987, many complex metal oxides have been found to possess super conductivity at some what higher temperature e.g.

$YBa_2Cu_3O_7$	90K
$Bi_2Ca_2Sr_2Cu_3O_{10}$	105K
$Ti_2Ca_2Ba_2Cu_3O_{10}$	125K

Super conductivity materials have great technical potentials. They can be used in electronics in building magnets, in power transmission and levitation transportation (trains which move in air without rails).

Illustration 49. *Out of $SiO_2(s)$, Si(s), NaCl(s) and $Br_2(l)$ which is the best electrical conductor?*

Solution: Si(s) because only this is a semi-conductor, while others $SiO_2(s)$, NaCl(s) and $Br_2(l)$ are insulators.

Illustration 50. *Explain:*

(a) The basis of similarities and difference between metallic and ionic crystals.

(b) Unit cell is not simply a cube of $4Na^+$ ions and $4Cl^-$ ions.

(c) Can a cube consisting of Na^+ and Cl^- ions at alternate corners serve as satisfactory unit cell for the sodium chloride lattice?

(d) Ionic solids are hard and brittle.

Solution:

(a) **Similarities:**

(i) Both involve electrostatic forces of attraction.

(ii) Both are non-directional.

Differences: Ionic bond is a strong bond due to electrostatic forces of attraction while metallic bond may be weak or strong depending upon the kernels.

(b) Unit cell of NaCl has fcc arrangement of Cl^- ions and Na^+ ions are present at the edge centres and one at the body-cnetre.

Thus there are $14Cl^-$ ions and $13Na^+$ ions in the unit cell. However their net contribution towards the unit cell is $4Na^+$ and $4Cl^-$ ions.

(c) Yes because its repetition in different directions produces the complete space lattice.

(d) Ionic solids are hard because there are strong electrostatic forces of attraction. However they are brittle because the bond is non-directional.

Illustration 51. *Analysis shows that nickel oxide has the formula $Ni_{0.98}O$. What fractions of the nickel exist as Ni^{2+} and Ni^{3+} ions?*

Solution: 98 Ni atoms are associated with 100 O-atom. Out of 98 Ni atoms, suppose Ni present as $Ni^{2+} = x$

Then Ni present as $Ni^{3+} = 98 - x$

Total charge on xNi^{2+} and $(98 - x)Ni^{3+}$ should be equal to charge n $100O^{2-}$. Hence

$x \times 2 + (98 - x) \times 3 = 100 \times 2$

or $2x + 294 - 3x = 200$

or $x = 94$

$\therefore$ Fraction of Ni present as $Ni^{2+} = \frac{94}{98} \times 100 = 96\%$

$\therefore$ Fraction of Ni present as $Ni^{3+} = \frac{4}{98} \times 100 = 4\%$

ANSWERS TO EXERCISES

Exercise 1:

One

Exercise 2:

(i) X atom is present at the every corner of cube and each corner atom makes a contribution of $\frac{1}{8}$

$\therefore$ No of X atom per unit cell = $\frac{1}{8} \times 8 = 1$

Y atoms are present only at one set of opposite faces and each atom at the face centre makes a contribution of $\frac{1}{2}$

$\therefore$ The no. of atoms of Y = $2 \times \frac{1}{2} = 1$

Thus, the formula of the compound is XY

(ii) In a face centred cubic cell, there are four atoms per unit cell.

The no of gold atoms in 2.0 g gold = $\frac{6.023 \times 10^{23}}{197} \times 2$

The no of unit cell = $\frac{6.023 \times 10^{23} \times 2}{197 \times 4} = 1.53 \times 10^{21}$

Exercise 3:

12

Exercise 4:

(a) 48%, 32%, 26%

(b) fcc

Exercise 5:

(a) (i) 100% (iii) 50%

(b) AB_2O_4

Exercise 6:

(i)

	Octahedral void	Tetrahedral void
HCP	6	12
FCC	4	8

(ii) False

Exercise 7:

(i) 4, 6, 8

(ii) 6, Octahedral

Exercise 8:

4

Exercise 9:

(i) r = 124 pm

(ii) $N_A = 6.07 \times 10^{23}$

Exercise 10:

105.65 gram

Exercise 11:

(i) 0.6256A°

(ii) 2.51A°

Exercise 12:

6 : 6 and 8 : 8

MISCELLANEOUS EXERCISES

Exercise 1: *How many atoms are present per unit cell in a primitive unit cell?*

Exercise 2: *Name two voids.*

Exercise 3: *What is the contribution of the atom when it is placed at the centre of the edge in a unit cell of a cube?*

Exercise 4: *What is the coordination number in a single plane of a closed packed structure?*

Exercise 5: *What is the effect on the density of a crystal due to Schottky defect?*

Exercise 6: *What is the atomic radius of the atom in body–centred cubic structure?*

Exercise 7: *How much space is empty in face–centred cubic structure?*

Exercise 8: *What is the coordination number in hcp and ccp arrangements?*

Exercise 9: *A given solid can belong to one of the four crystal types, i.e., ionic, molecular, covalent and metallic. Indicate the crystal type of the following:*

(a) Diamond
(b) Sodium chloride
(c) Ice
(d) Copper
(e) Boron nitride
(f) Zinc oxide
(g) Paraffin wax

Exercise 10: *What types of crystals would be?*

(a) The hardest
(b) The softest
(c) The highest melting
(d) The lowest melting
(e) Electrically conducting

ANSWER TO MISCELLANEOUS EXERCISES

Exercise 1: One atom per unit cell

Exercise 2: Tetrahedral and octahedral

Exercise 3: 0.25

Exercise 4: 6

Exercise 5: Density decreases

Exercise 6: $\frac{\sqrt{3}}{4}$a where a is the length of the edge

Exercise 7: 26%

Exercise 8: 12

Exercise 9:
(a) Covalent
(b) Ionic
(c) Molecular
(d) Metallic
(e) Covalent
(f) Ionic
(g) Covalent

Exercise 10:
(a) Covalent
(b) Molecular
(c) Covalent
(d) Molecular
(e) Metallic

SOLVED PROBLEMS

Subjective:

Board Type Questions

Prob 1. *How can you convert NaCl structure into CsCl structure and vice-versa?*

Sol. NaCl structure can be converted into CsCl structure by application of pressure while reverse can be done by heating to 760 K.

Prob 2. *AgI crystallizes in cubic close packed ZnS structure. What fraction of tetrahedral sites are occupied by Ag^+ ions?*

Sol. In the face-centred unit cell, there are eight tetrahedral voids, of these, half are occupied by silver cations.

Prob 3. *What is Frenkel defect?*

Sol. When some ions (usually cations) are missing from the lattice sites and they occupy the interstitial sites so that electrical neutrality as well as stoichiometry is maintained, it is called Frenkel defect.

Prob 4. *What type of crystal defect is produced when sodium chloride is doped with $MgCl_2$?*

Sol. It is called impurity defect. A cations vacancy is produced. A substitutional solid solution is formed (because $2Na^+$ ions are replaced by one Mg^{++} ion). This defect is also known as metal deficiency defect.

Prob 5. *A compound AB_2 possesses the CaF_2 type crystal structure. Write the co-ordination number of A^{++} and B^- ions in its crystals.*

Sol. Co-ordination number of A = 8
Co-ordination number of B = 4

IIT Level Questions

Prob 6. *A solid between A and B has the following arrangement of atoms*
(i) Atoms A are arranged in ccp array
(ii) Atoms B occupy all the octahedral voids and half the tetrahedral voids. What is the formula of the compound?

Sol. In a close packing, the number of octahedral voids is equal to the number of atoms and the number of tetrahedral voids is twice the number of atoms
Since all the octahedral voids and half the tetrahedral voids are filled there will be one atom of B in tetrahedral void and one atom in octahedral void corresponding to each A. Thus, there will be two atoms of B corresponding to each A.
Hence, formula of the solid is AB_2

Prob 7. *In corundum, oxide ions are arranged in hcp array and the aluminium ions occupy two thirds of octahedral voids. What is the formula of corundum?*

Sol. In ccp or hcp packing there is one octahedral void corresponding to each atom constituting the close packing. In corundum only $2/3^{rd}$ of the octahedral voids are occupied. It means corresponding to each oxide are 2/3 aluminium ions. The whole number ratio of oxide and aluminium ion in corundum is therefore 3:2 Hence formula of corundum is Al_2O_3

Prob 8. *Calculate the ratio of the alkali metal bromides on the basis of the data given below and predict the form of the crystal structure in each case. Ionic radii (in pm) are given below*
Li^+ = 74, Na^+ = 102, K^+ = 138
Rb^+ = 148, Cs^+ = 170, Br^- = 195

Sol. The ratio of cation to that of anion i.e. $\frac{r^+}{r^-}$ gives the clue for crystal structure

$$\frac{Li^+}{Br^-} = \frac{74}{195} = 0.379 \quad \text{(tetrahedral)}$$

$$\frac{Na^+}{Br^-} = \frac{102}{195} = 0.523 \quad \text{(octahedral)}$$

$$\frac{K^+}{Br^-} = \frac{138}{195} = 0.708 \quad \text{(octahedral)}$$

$$\frac{Rb^+}{Br^-} = \frac{148}{195} = 0.759 \quad \text{(Body centered)}$$

$$\frac{Cs^+}{Br^-} = \frac{170}{195} = 0.872 \quad \text{(Body centered)}$$

Prob 9. *In the close packed cation in an AB type solid have a radius of 75 pm, what would be the maximum and minimum sizes of the anions filling the voids?*

Sol. For close packed AB type solid

$$\frac{r^+}{r^-} = 0.414 - 0.732$$

$$\therefore \text{ Minimum value of } r^- = \frac{r^+}{0.732} = \frac{75}{0.732} = 102.5 \text{ pm}$$

$$\text{Maximum value } r^- = \frac{r^+}{0.414} = \frac{75}{0.414} = 181.2 \text{ pm}$$

Prob 10. *NH_4Cl crystallizes in a body centered cubic lattice, with a unit cell distance of 387 pm. Calculate (a) the distance between the oppositely charged ions in the lattice, and (b) the radius of the NH_4^+ ion if the radius of the Cl^- ion is 181 pm.*

Sol. (a) In a body centered cubic lattice oppositely charged ions touch each other along the cross - diagonal of the cube. Hence, we can write,

$$2r^+ + 2r^- = \sqrt{3}a, \quad r^+ + r^- = \frac{\sqrt{3}a}{2} = \frac{\sqrt{3}}{2}(387\,\text{pm}) = 335.15\,\text{pm}$$

(b) Now, since $r^- = 181\text{pm}$,
we have $r^+ = (335.15 - 181)\text{pm} = 154.15\,\text{pm}$.

Prob 11. *Copper has the fcc crystal structure. Assuming an atomic radius of 130pm for copper atom (Cu = 63.54):*
(a) What is the length of unit cell of Cu?
(b) What is the volume of the unit cell?
(c) How many atoms belong to the unit cell?
(d) Find the density of Cu.

Sol. As we know

$$\rho = \frac{n \times M_m}{N_A \times a^3},$$

(a) for fcc structure

$4r = \sqrt{2}\, a$

$a = 2\sqrt{2}\, r$

$= 2\sqrt{2} \times 130$ pm = 367.64 pm

(b) volume of unit cell = $a^3 = \left(367.64 \times 10^{-10}\,\text{cm}\right)^3$

$= 4.968 \times 10^{-23}\ \text{cm}^3$

(c) n = 4

(d) $\rho = \dfrac{4 \times 63.54}{6.023 \times 10^{23} \times (3.67 \times 10^{-8}\,\text{cm}^3)^3} = 8.54\ \text{gm / cm}^3$

Prob 12. *The density of CaO is 3.35 gm/cm³. The oxide crystallises in one of the cubic systems with an edge length of 4.80 Å. How many Ca^{++} ions and O^{-2} ions belong to each unit cell, and which type of cubic system is present?*

Sol. From equation

ρ(density) = 3.35 gm/cm³

a = 4.80 Å

M_m of CaO = (40 + 16) gm = 56 gm CaO

$\because \rho = \dfrac{n \times M_m}{a^3 \times N_A}$ where n = no. of molecules per unit cell

$\therefore n = \dfrac{3.35 \times (4.8 \times 10^{-8})^3 \times 6.023 \times 10^{23}}{56} = 3.98$

or $n \approx 4$

So, 4-molecules of CaO are present in 1 unit cell

So, no. of Ca^{++} ion = 4

No. of O^{--} ion = 4

So, cubic system is fcc type.

Prob 13. *A metal crystallizes into two cubic system-face centred cubic (fcc) and body centred cubic (bcc) whose unit cell lengths are 3.5 and 3.0Å respectively. Calculate the ratio of densities of fcc and bcc.*

Sol. fcc unit cell length = 3.5Å

bcc unit cell length = 3.0Å

$\text{Density in fcc} = \dfrac{n_1 \times \text{atomic weight}}{V_1 \times \text{Avogadro number}}$

$\text{Density in bcc} = \dfrac{n_2 \times \text{atomic weight}}{V_2 \times \text{Avogardro number}}$

$\dfrac{D_{fcc}}{D_{bcc}} = \dfrac{n_1}{n_2} \times \dfrac{V_2}{V_1}$

n_1 for fcc = 4; Also $V_1 = a^3 = (3.5 \times 10^{-8})^3$

n_2 for bcc = 2; Also $V_2 = a^3 = (3.0 \times 10^{-8})^3$

$\dfrac{D_{fcc}}{D_{bcc}} = \dfrac{4 \times (3 \times 10^{-8})^3}{2 \times (3.5 \times 10^{-8})^3} = 1.259$

Prob 14. *Copper crystal has a face centred cubic structure. Atomic radius of copper atom is 128 pm. What is the density of copper metal? Atomic mass of copper is 63.5.*

Sol. In face centred cubic arrangement face diagonal is four times the radius of atoms face diagonal = $4\times128 = 512$ pm

Face diagonal = $\sqrt{2}\times$ edge lengh

Edge length = $\frac{512}{\sqrt{2}} = 362$ pm = 362×10^{-10} cm

Volume of the unit cell = $(362\times10^{-10})^3$ cm^3 = 47.4×10^{-24} cm^3

In a face centred cubic unit cell, there are four atoms per unit cell

Mass of unit cell = $\frac{4\times63.5}{6.023\times10^{23}}\text{g} = 4.22\times10^{-22}\text{g}$

Density = $\frac{\text{mass of unit cell}}{\text{volume of unit cell}} = \frac{4.22\times10^{-22}}{47.4\times10^{-24}} = 8.9\ \text{g cm}^{-3}$

Prob 15. *The first order reflections of a beam of X – rays of wavelength of 1.54A° from the (100) face of a crystal of the simple cubic type occurs at an angle 11.29°. Calculate the length of the unit cell.*

Sol. Applying Bragg's equation

$2\ d\ \text{Sin}\theta = n\lambda$

Given $\theta = 11.29°$, $\lambda = 1.54\text{A}° = 1.54\times10^{-8}$ cm

$n = 1$

$d = \frac{1.54\times10^{-8}}{2\times\text{Sin}11.29°} = \frac{1.54\times10^{-8}}{2\times0.1957} = 3.93\times10^{-8}\text{cm}$

$d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}} = a$

$a = 3.93\times10^{-8}$ cm (length of the unit cell)

Prob 16. *X- rays of wavelength equal to 0.134 nm give a first order diffraction from the surface of a crystal when the value of θ is 10.5°. Calculate the distance between the planes in the crystal parallel to the surface examined.*

Sol. Given $\lambda = 0.134$ nm, $\theta = 10.5°$

$n = 1$

Applying Bragg's equation

$2d\ \text{Sin}\theta = n\lambda$

$d = \frac{n\lambda}{2\text{Sin}\theta} = \frac{1\times0.134}{2\times\text{Sin}10.5°} = \frac{0.134}{2\times0.1822} = 3.68\ \text{Å}$

Prob 17. *What is the difference in the semi conductors obtained by doping silicon with Al or with P?*

Sol. Silicon doped with Al produces P – type semi conductors i.e. flow is due to creation of positive holes whereas silicon doped with P produces n – type semi conductors i.e. flow is due to extra electrons carrying negative charge.

Prob 18. *Non stoichiometric cuprous oxide, Cu_2O can be prepared in the laboratory. In this oxide, copper to oxygen ratio is slightly less than 2:1 can you account of the fact that this substance is a p – type semiconductor*

Sol. The ratio less that 2:1 in Cu_2O show that some cuprous (Cu^{+}) ions have been replaced by cupric (Cu^{+2}) ions. To maintain electrical neutrality, every two Cu^{+} ions will be replaced by one Cu^{+2} ion. Thereby creating a hole. As conduction will be due to presence of these positive holes, hence it is a p – type semi conductor.

Prob 19. *Classify each of the following as being either a p – type or n – type semiconductor.*
(i) Ge doped with In
(ii) B doped with Si

Sol. (i) Ge is group 14 element and In is group 13 element. Hence an electron deficient hole is created and therefore, it is p – type.
(ii) B is group 13 elements and Si is group 14 elements, there will be a free electron. Henc, it is n – type

Prob 20. *If NaCl is doped with 10^{-3} mole% $SrCl_2$ what is the concentration of cation vacancies?*

Sol. Doping of NaCl with 10^{-3} mol% $SrCl_2$ means that 100 moles of NaCl are doped with 10^{-3} mol of $SrCl_2$

$\therefore$ 1 mole of NaCl is doped with $SrCl_2 = \frac{10^{-3}}{100}\text{mole} = 10^{-5}$ mole

As each Sr^{+2} ion creates one cation vacancy, therefore concentration of cation vacancies
$= 10^{-5}$ mol / mol of NaCl
$=10^{-5}\times6.023\times10^{23} = 6.023\times10^{18}$ mol^{-1}

Objective:

Prob 1. *If the radius ratio is in the range of 0.414 – 0.732, then the coordination number will be*

(A) 2 *(B) 4*

(C) 6 *(D) 8*

Sol. (C) for radius ratio in the range 0.414 – 0.732 coordination number = 6

Prob 2. *If the distance between Na^+ and Cl^- ions in NaCl crystal is 'a' pm, what is the length of the cell edge?*

(A) 2 a pm *(B) a/2 pm*

(C) 4 a pm *(D) a/4 pm*

Sol. Length of the edge of NaCl unit = 2×distance between Na^+ & Cl^- = 2×a

$\therefore$ (A)

Prob 3. *The coordination number of each atom in bcc is*

(A) 4 *(B) 6*

(C) 8 *(D) 12*

Sol. The coordination number of each atom in bcc is 8

$\therefore$ (C)

Prob 4. *If R is the radius of the spheres in the close packed arrangement and 'r' is the radius of the octahedral void, then*

(A) R = 0.414 r *(B) r = 0.414 R*

(C) R = 0.225 r *(D) r = 0.224 R*

Sol. For the octahedral void $\frac{r_{void}}{r_{sphere}} = 0.414$ or $\frac{r}{R} = 0.414$ or r = 0.414 R

$\therefore$ (B)

Prob 5. *The coordination number of a cation occupying a tetrahedral hole and an octahedral hole respectively are*

(A) 4,6 (B) 6,4

(C) 8.4 (D) 4, 8

Sol. The coordination of a cation occupying tetrahedral hole is 4 while that of a cation occupying an octahedral void is 6

$\therefore$ (A)

Prob 6. *LIF is an example of*

(A) Ionic crystal *(B) Metallic crystal*

(C) Covalent crystal *(D) Molecular Crystal*

Sol. LIF is an ionic solid

$\therefore$(A)

Prob 7. *Which of the following adopts normal spinel structure?*

(A) CsCl (B) $MgAl_2O_4$

(C) FeO (D) CaF_2

Sol. $MgAl_2O_4$ has a spinel structure

$\therefore$(B)

Prob 8. *The electricity produced on applying stress on the crystal is called*

(A) Pyroelectricity *(B) Piezoelectricity*

(C) Ferro electricity *(D) Anti – Ferro electricity*

Sol. Electricity produced on applying stress is called piezoelectricity

$\therefore$(B)

Prob 9. *Ferromagnetism is maximum in*

(A) Fe *(B) Ni*

(C) Co *(D) none*

Sol. Ferrimagnetism is maximum in Fe out of Fe, Co & Ni

$\therefore$(A)

Prob 10. *Most crystal show good cleavage because their atoms, ions or molecules are*

(A) Weakly bonded together *(B) Strongly bonded together*

(C) Spherically symmetrical *(D) Arranged in planes*

Sol. Crystal show good cleavage because the their constituent particles are arranged in planer

$\therefore$(D)

Prob 11. *If we mix a pentavalent impurity in a crystal lattice of germanium, what type of semiconductor formation will occur?*

(A) P – type *(B) n – Type*

(C) Both P & n type *(D) none of these*

Sol. Pentavalent impurity mixed in germanium gives n – type semi conductor.

$\therefore$(B)

Prob 12. *A solid is made of two elements P & Q. Atoms P are in ccp arrangements and atoms Q occupy all the octahedral voids and half of the tetrahedral voids, then the simples formula of the compound is*

(A) PQ_2 *(B)* P_2Q

(C) PQ *(D)* P_2Q_2

Sol. Four atoms (P) contributes to one unit cell from ccp arrangement and 4 – atoms (Q) from the all octahedral voids and 4 – atoms (Q) from the half of the tetrahedral void contributes one unit cell. So formula of solid is P_4Q_8, so the simplest formula of the solid is PQ_2.

$\therefore$(A)

Prob 13. *Xenon crystallizes in fcc lattice and the edge of the unit cell is 620 pm, then the radius of xenon atom is*

(A) 438.5 pm *(B) 219.25 pm*

(C) 536.94 pm *(D) 265.5 pm*

Sol. For fcc lattice $4r = \sqrt{2} \times a$, where a = 620 pm

or $r = \frac{1}{2\sqrt{2}} a$

$\frac{1}{2\sqrt{2}} \times 620\text{pm} = 219.20\text{pm}$

$\therefore$(B)

Prob 14. *An ionic compound AB has ZnS type of structure, if the radius A^+ is 22.5 pm then the ideal radius of B^- is*

(A) 54.35 pm *(B) 100 pm*

(C) 145.16 pm *(D) None*

Sol. Since ionic compound AB has ZnS type of structure, therefore it has tetrahedral holes for which $\frac{\text{radius of cation}}{\text{radius of anion}} = 0.225$

$\frac{r^+}{r^-} = 0.225 \Rightarrow \frac{22.5}{r^-} = 0.225$

Hence r^- = 100 pm

$\therefore$ (B)

Prob 15. *A metal M has fcc arrangement and edge length of the unit cell is 400 pm. The atomic radius of 'M' is*

(A) 100 pm *(B) 200 pm*

(C) 141 pm *(D) 173 pm*

Sol. In fcc, the corners and face sphere touch each other

$\therefore$ The diagonal length = $\sqrt{2}a$

$\sqrt{2}a = 4r$

$r = \frac{\sqrt{2}}{4} \times 400 = \sqrt{2} \times 100 = 1.414 \times 100$ = 141.4 pm

$\therefore$(D)

Fill in the Blanks

Prob 16. Amorphous solids …… have …….. melting points.

Sol. do not, sharp

Prob 17. A crystal is defined as a solid figure which has a definite geometrical shape with ……. faces and ……. edges.

Sol. flat, sharp

Prob 18. In NaCl ionic crystal, each Na^+ ion is surrounded by …….. Cl^- ions and each Cl^- ion is surrounded by…….. Na^+ ions.

Sol. six, six

Prob 19. In cubic crystal, there are …… elements of symmetry.

Sol. 23

Prob 20. The crystal structure of CsCl is ……..

Sol. body–centred cubic

True and False

Prob 21. Liquid and solid states are the condensed states of matter.

Sol. False

Prob 22. Volatile liquids have high cohesive forces.

Sol. False

Prob 23. Viscosity decreases with decrease in temperature.

Sol. True

Prob 24. Crystalline solids are anisotropic.

Sol. True

Prob 25. A crystal possesses only one centre of symmetry.

Sol. True

ASSIGNMENT PROBLEMS

Subjective:

Level – 0

1. Stability of a crystal is reflected in the magnitude of its melting point. Comment

2. The ions of NaF and MgO all have the same number of electrons and internuclear distances are about the same (235 pm and 215 pm), why then are the melting points of NaF and MgO, so different (992°C and 2642°C)?

3. What happens when Ferromagnetic or anti Ferromagnetic or a Ferrimagnetic solid is heated?

4. What is the difference between glass and Quartz while both are made up from SiO_4 tetrahedral?
Under what conditions could Quartz be converted into glass?

5. Why common salt is sometimes yellow instead of being pure white?

6. Why is Frenkel defect not found in pure alkali metal halides?

7. Why the defects of the crystalline solids are called thermodynamic defects?

8. Why stoichiometric defects are also called instrinsic defects?

9. What is radius ratio? What is its significance?

10. State the difference between Schottky and Frenkel defects? Which of these two changes the density of the solid?

11. The unit cube length of LiCl (NaCl structure) is 5.14A°. Assuming anion-anion contact, calculate the ionic radius of Cl^- ion.

12. Ferric oxide crystallizes in a hexagonal close – packed array of oxide (O^{2-}) ions with two out of every three octahedral holes occupied by iron ions. What is the formula of ferric oxide?

13. A metal crystallises into two cubic phases, face – centered cubic (fcc) and body – centred cubic (bcc), whose unit cell lengths are 3.5 and 3.0 $\mathring{A}$, respectively. Calculate the ratio of density of fcc and bcc.

14. The edge length of a cubic unit cell of an element (atomic mass=95.54) is 313 pm and its density is 10.3 g/ml calculate the atomic radius.

15. Xenon crystallizes in face centre cubic lattice and the edge of the unit cell is 620 pm, then calculate the radius of Xenon-atom.

Level - I

1. In a face centred lattice of X and Y, X atom are present at the corners while Y atoms at the face centres.
(a) What is the formula of compound?
(b) What would be the formula of the compound if
(i) Two atoms of X are missing from the corner.
(ii) One of the X atoms is missing from a corner is replaced by Z atoms(also monovalent).

2. Lithium metal has a bcc structure. Its density is 0.53 g cm^{-3} and its molar mass is 6.94 g mol^{-1}. Calculate the volume of a unit cell of lithium metal.

3. If three elements P, Q and R crystallises in a cubic solid lattice with P atoms at the corners, Q atoms at the cube centres and R atoms at the centre of the face of the cube, then write the formula of the compound.

4. A metallic element 'X' exists as a cubic lattice. Each edge of the unit cell is 2.90 A° and density of metal is 7.20 g/cm^3. How many unit cells will be present in 100 g of the metal?

5. Ferric oxide crystallises in a hexagonal close packed array of oxide ions with two out of every three octahedral holes occupied by Ferric ions. Derive the formula of the Ferric oxide.

6. In the compound AX, the radius of A^+ ion is 95 pm and that of X^- ion is 181 pm. Predict the crystal structure of AX and write the coordination number of each of the ion.

7. A solid A^+ B^- has a NaCl type close packed structure. If the anion has a radius of 250 pm. What should be the ideal radius of the cation? Can a cation C^+ having a radius of 180 pm be slipped into the tetrahedral site of the crystal A^+B^-? Give reason for your answer.

8. How can you determine the atomic mass of an unknown metal, if you know its density and the dimension of its unit cell? Explain your answer

9. Copper crystallises into fcc lattice with edge length 3.61×10^{-8} cm. Show that the calculated density is in agreement with its measured value of 8.92 g cm^{-3}.

10. What will be the wavelength of the X – rays which give a diffraction angle 2θ equal to 16.80° for a crystal, if the inter planar distance in the crystal is 0.200 nm and only first order diffraction is observed (sin 8.40° = 0.146)

Level – II

1. A compound formed by elements X and Y has cubic unit cells in which X – atoms are at the corners of the cube and Y – atoms are at the face centre. What is the formula of the compound? If the atomic weight of X and Y are 10 and 15 respectively, then find out the no of unit cell present in 20 gm of compound.

2. A solid AB has CsCl type structure. The edge length of the unit cell is 404 pm. Calculate the distance of closest approach between A^+ and B^- ions.

3. If the radius of bromide ion is 0.182 nm. How large a cation can fit in the tetrahedral hole?

4. The compound CuCl has ZnS structure and the edge length of its unit cell is 500 pm. Calculate its density (At. Mass Cu = 63, Cl = 35.5)

5. CsCl has cubic structure, Its density is 3.99 g/cm^3. What is the distance between Cs^+ and Cl^- ions (At mass of Cs = 133)

6. Calculate the value of Avogadro's number from the following data:
Density of KF = 2.48 g cm^{-3}.
Distance between K^+ and F^- in KF = 269 pm (At mass K = 39, F = 19 amu)

7. Thallium chloride, TlCl crystallise in either a simple cubic lattice or a face centred cubic lattice of Cl^- ions with Tl^+ in the holes. If the density of the solid is 9.00 g cm^{-3} and edge of the unit cell is 3.85×10^{-8} cm. What is the unit cell geometry?

8. Formula mass of NaCl is 58.45 g mol^{-1} and density of its pure form is 2.167 g cm^{-3}. The average distance between adjacent sodium and chloride ions in the crystal is 2.814×10^{-8} cm. Calculate Avogadro's constant.

9. An element crystallises in a structure having fcc unit cell of an edge 200 pm. Calculate the density if 200 g of this element contains 24×10^{23} atom.

10. Analysis shows that iron oxide has formula $Fe_{0.93}O_{1.00}$. What fractions of the iron exist as Fe^{+2} and Fe^{+3} ions?

Objective:

Level – I

1. If the coordination no. of an element in its crystal lattice is 8, then packing is
(A) fcc (B) hcp
(C) bcc (D) none of the above

2. In a hexagonal closest packing in two layers one above the other, the coordination number of each sphere will be
(A) 4 (B) 6
(C) 8 (D) 9

3. The maximum proportion of available volume that can be filled by hard spheres in diamond is
(A) 0.52 (B) 0.34
(C) 0.32 (D) 0.68

4. The number of molecules in a unit cell of fluorite is
(A) 2 (B) 4
(C) 6 (D) 8

5. 4:4 Co – ordination is found in
(A) ZnS (B) CuCl
(C) AgI (D) Au

6. Which one of the following is a Ferrite?
(A) $Na_2Fe_2O_4$ (B) $MgFe_2O_4$
(C) $AlFe_4O_4$ (D) Zn_3FeO_4

7. When NaCl is doped with $MgCl_2$, the nature of defect produced is
(A) Interstitial defect (B) Schottky defect
(C) Frenkel defect (D) None of these

8. To get n – type doped Semi conductor, impurity to be added to silicon should have the following number of valence electrons
(A) 2 (B) 5
(C) 3 (D) 1

9. Super conductors are derived from compounds of
(A) p – block (B) Lanthanides
(C) Actinides (D) Transition elements

10. At zero Kelvin, most of the ionic crystals possess
(A) Frenkel defect (B) Schottky defect
(C) Metal excess defect (D) No defect

11. Non – stoichiometric metal deficiency is shown in the salts of

(A) All metals
(B) Alkali metals
(C) Alkaline earth metals
(D) Transition metal

12. Silicon doped with arsenic is

(A) p – type Semiconductor
(B) n – type Semiconductor
(C) Like a metallic conductor
(D) an insulator

13. A solid has a structure in which W atoms are located at the corners of cubic lattice, O atoms at the centre of edges and Na atom at the centre of the cube. The formula of the compound is

(A) $NaWO_2$
(B) $NaWO_3$
(C) Na_2WO_3
(D) $NaWO_4$

14. In a face centred cubic arrangement of A and B atoms in which A atoms are at the corners of the unit cell and B atoms at the face centres, one of the A atoms is missing from one corner in unit cell. The simplest formula of compound is

(A) A_7B_3
(B) AB_3
(C) A_7B_{24}
(D) $A_{7/8}B_3$

15. A binary solid (A^+B^-) has a rock salt structure. If the edge length is 400 pm and radius of cation is 75 pm. The radius of anion is

(A) 100 pm
(B) 125 pm
(C) 250 pm
(D) 325 pm

Level – II

1. A binary solid (A^+ B^-) has zinc blende structure with B^- ions constituting the lattice and A^+ ions occupying 25% tetrahedral holes. The formula of solid is
(A) AB (B) A_2B
(C) AB_2 (D) AB_4

2. In a metal oxide, the oxide ions are arranged in hexagonal close packing and metal ions occupy two third of the octahedral voids. The formula of the oxide is
(A) Mo (B) M_2O_3
(C) Mo_2 (D) M_2O

3. The radius of A^+ is 95 pm and that of B^- ion is 181 pm. Hence the coordination number of A^+ will be
(A) 4 (B) 6
(C) 8 (D) Un predictable

4. The empty space in hexagonal close packing pattern is
(A) 48% (B) 52.4 %
(C) 26% (D) 70%

5. Which of the following crystallises in bcc structure?
(A) ZnS (B) Na_2S
(B) NaCl (D) CsCl

6. The structure of Na_2O crystal is
(A) CsCl type (B) NaCl type
(C) ZnS type (D) Anti fluorite

7. A solid AB has NaCl type structure. If the radius of the cation A is 100 pm, then the radius of the anion B will be
(A) 241 pm (B) 414 pm
(C) 225 pm (D) 44.4 pm

8. Doping of silicon with P or Al increasing the conductivity. The difference in the two cases is
(A) P is non – metal whereas Al is a metal
(B) P is a poor conductor while Al is conductor
(C) P gives rise to extra electrons while Al gives rise to holes
(D) P gives rise to holes while Al gives rise to extra electrons

9. In the laboratory, sodium chloride is made by burning sodium in the atmosphere of chlorine. The salt obtained is yellow in colour. The cause of yellow colour is due to
(A) Presence of Na^+ ions in the crystal lattice
(B) Presence of Cl^- ions in the crystal lattice
(C) Presence of electrons in the crystal lattice
(D) Presence of faced centred cubic crystal lattice

10. The C – C and Si – C interatomic distances are 154 pm and 188 pm. The atomic radius of Si is

(A) 77 pm (B) 94 pm

(C) 114 pm (D) 111 pm

11. ZnO is white when cold and yellow when heated, it is due to the development of

(A) Frenkel defect (B) Schottky defect

(C) Metal excess defect (D) Metal deficiency defect

12. The second order Bragg diffraction of X – Rays with λ = 1.0A° from a set of parallel planes in a metal occurs at an angel of 60°. The distance between the Scattering planes in the crystal is

(A) 0.575 A° (B) 1.00 A°

(C) 2.00 A° (D) 1.15 A°

13. A mineral having the formula AB_2 crystallizes in the cubic close packed lattice, with the A atom occupying the lattice points. The Co – ordination number of B atoms and the fraction of the tetrahedral sites occupied by B atoms are

(A) 4,100% (B) 6,75%

(C) 1, 25% (D) 6,50%

14. A crystal of lead (II) sulphide has NaCl structure. In this crystal the shortest distance between a Pb^{+2} ion and S^{-2} ion is 297 pm. What is the volume of unit cell in lead sulphide?

(A) 209.6×10^{-24} cm^3 (B) 207.8×10^{-23} cm^3

(C) $22.3x10^{-23}$ cm^3 (D) 209.8×10^{-23} cm^3

15. How many unit cells are present in a cube shaped ideal crystal of NaCl of mass 1.00 g?

(A) 5.14×10^{21} (B) 1.28×10^{21}

(C) 1.71×10^{21} (D) 2.57×10^{21}

ANSWERS TO ASSIGNMENT PROBLEMS

Subjective:

Level - O

1. Higher the melting point greater are the forces holding the constituent particles together and hence greater is the stability.

2. In Mg^{+2} O^{-2} both the ions carry two units of charge whereas in $Na^{+}F^{-}$, each ion carries only one unit of charge. Hence due to greater charge density, electrostatic forces of attraction in MgO are stronger.

3. It changes into paramagnetic at some temperature.

4. Glass is amorphous solid whereas Quartz is a crystalline solid. On melting Quartz and then rapidly cooling it, Quartz is converted into glass.

5. This is due to presence of electron in some lattice sites in place of anions. These sites act as F – centres.

6. This is due to the fact that alkali metal ions have large size which cannot fit into the interstitial sites.

7. There is perfect arrangement of the constituent particles only at zero Kelvin. As the temperature increases (say upto room temperature) the chance that a lattice sites may be unoccupied by an ion increases. As the number of defect increases with temperature the defects called thermodynamic defects.

8. As they do not alter the stoichiometry of the crystal

11. The distance between Li^{+} and Cl^{-} ion can be derived as half of the edge length of cube

$2r_{+} + 2r_{-} = 5.14$

$4r_{-} = \sqrt{3} \times 5.14$

$\therefore \frac{r_{+}}{r_{-}} = 0.414$

$0.414\ r_{-} + r_{-} = 2.57$

$r_{-} = 1.817$

12. Number of oxide ions per unit cell = 6

$\therefore$ Number of octahedral voids = 6

Since two out of every three octahedral voids are occupied by iron

$\therefore$ number of iron ions per nit cell = $\frac{2}{3} \times 6 = 4$

$\therefore$ formula is Fe_4O_6 or Fe_2O_3

13. 1.259

14. $e = \frac{Z \times M}{N \times a^3}$

$Z = \frac{10.3 \times 6.02 \times 10^{23} \times (313)^3 \times 10^{-30}}{95.54}$

$Z \approx 2$

Therefore the unit cell is b.c.c

$\therefore 4r = \sqrt{3}\ a$

$r = \frac{\sqrt{3}}{4}\ 313 = 135.5$ pm

15. For fcc lattice

$4r = \sqrt{2}a$ where a = 620 pm

or, $r = \frac{1}{2\sqrt{2}} \times a$

$\frac{1}{2\sqrt{2}} \times 620$ pm = 219.20 pm

Level – I

1. (a) XY_3 (b) (i) XY_4 (ii) $X_7Y_{24}Z$

2. 4.35×10^{-23} cm^3

3. PQR_3

4. 5.7×10^{23}

5. Fe_2O_3

6. $r^+/r^- = 0.524 \therefore$ 6, octahedral

7. 103.4 pm, No

9. Calculate value = 8.96 g/cm^3, Measured value = 8.92 g/cm^3, $\therefore$ densities are nearly identical.

10. 5.84×10^{-11}

Level – II

1. XY_3, 2.19×10^{23}

2. 349.9 pm

3. 0.041 nm

4. 5.24 g/cm^3

5. 357 pm

6. 6.0×10^{23}

7. $Z = 1.29 \approx 2$ (bcc)

8. 6.052×10^{23} mol^{-1}

9. 41.67 g cm^{-3}

10. Fe^{+2} = 84.94% and Fe^{+3} = 15.06%

Objective:

Level - I

1.	**C**	2.	**D**	3.	**B**
4.	**B**	5.	**A**	6.	**B**
7.	**D**	8	**B**	9.	**D**
10.	**D**	11.	**D**	12.	**B**
13.	**B**	14.	**C**	15.	**B**

Level - II

1.	**C**	2.	**B**	3.	**B**
4.	**C**	5.	**D**	6.	**D**
7	**A**	8.	**C**	9.	**C**
10.	**D**	11.	**C**	12.	**D**
13.	**A**	14.	**A**	15.	**D**